

Modest and politically contingent stock market value for corporate climate pledges

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Whether financial markets reward or penalize voluntary corporate decarbonization pledges shapes the private sector’s incentive to decarbonize. We assemble a global dataset of 558 time-stamped climate-commitment press releases by listed firms (2017–2023), sampled from a financial newswire to maximize the detectability of any market reaction. Announcements raise stock prices by a modest 0.25% on average (95% confidence interval 0.05–0.46%) over the first three trading days, after which the effect fades. Reactions are statistically indistinguishable across third-party certification labels and are significant only for US-listed firms. They do not increase with carbon-price stringency; if anything, they are largest where no carbon price applies. Within the United States, the anti-ESG backlash is associated with a relative two-percentage-point decline in announcement returns for exposed firms, alongside a sharp fall in those firms’ share of global announcements. The financial incentive for public climate commitment is real but modest, short-lived and politically contingent.

The intensification of climate policy and the associated “transition risks” have catalysed a global surge in voluntary corporate greenhouse-gas (GHG) mitigation targets. Whether financial markets actually reward firms for making these pledges matters for the private sector’s role in decarbonization. Investors increasingly demand disclosure of firms’ climate exposure [1], and because equity prices are forward-looking, the market response to a pledge aggregates investors’ beliefs about the costs, benefits and credibility of the announced transition path, and thereby shapes the incentive to commit, and to commit publicly, in the first place [2].

The sign of this response is theoretically ambiguous. A credible commitment can raise firm value by reducing exposure to transition risk, lowering the cost of capital, attracting demand for green assets [3] or reducing the cost of credit through the bank-lending channel

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[4]. But credible commitments require costly capital expenditure, and furthermore corporate pledges often outrun action [5–7], while pledges perceived as cheap talk may be discounted or even penalized.

Existing evidence links carbon risk to equity returns [8], measures whether announced targets have a transformative impact on the corporate environmental performance [9–12], discusses their skepticism about the credibility of these targets [13–16], tracks whether markets care about their eventual outcome [17], and finally studies long-horizon performance of committed firms portfolios [18–20]. Direct evidence on how markets price the announcement itself is far scarcer and largely confined to the United States or to single industries. Event studies of eco-friendly initiatives and green-bond issues find positive reactions [21, 22], sustainability-linked bonds now tie financing costs directly to such targets [23], US corporate “green pledges” identified from news text earn positive and persistent announcement returns [24], and net-zero pledges by US oil and gas producers can be penalized unless backed by credible governance arrangements [25]. Broader evidence has been held back in good part because commitment databases record adoption dates imprecisely, blurring the moment at which the information reaches investors.

We address three open questions. Do markets place a value on the public act of commitment at the moment it is disclosed? Does any such value reflect the credibility conferred by third-party certification – a natural conjecture if validation separates genuine pledges from cheap talk [26]? And is it contingent on the policy and political environment in which the firm operates [27, 28]?

To answer them, we assemble a global dataset of time-stamped corporate climate-commitment press releases from the LSEG (ex-Refinitiv) press aggregator. This sampling choice is deliberate and central to our interpretation. LSEG financial-press wire is a feed closely monitored by financial market participants, and by dating each release to the day it became public, cross-checking each timestamp against public sources, we restrict attention to pledges that demonstrably reached investors as salient, sharply timed news rather than disclosures buried in sustainability reports. This is the most favorable setting in which to detect a market reaction: any attenuation from information failing to reach investors, or from imprecise timing, works against finding an effect, so a near-zero, short-lived premium measured here is unlikely to be an artefact of weak or mistimed information.

A keyword search returns 947 candidate releases over 2017–2023, of which 716 are press releases by listed firms with usable equity identifiers. After the event-study data-preparation pipeline (Methods), the analysis sample comprises 558 announcements for which we have at least one exploitable abnormal return. We estimate the stock-market reaction to these announcements with a daily event study: for each firm, we compare the realized stock

return in the days around the announcement with the return predicted by the firm’s usual co-movement with its home-market index; the difference - the abnormal return - captures the value investors attach to the news, and averaging it across events isolates the market’s response to the announcements themselves. We then examine how reactions vary with commitment characteristics, certification labels, sectors, and the policy and political environment.

Three findings emerge. First, the average market reaction is positive but small and short-lived: cumulative average abnormal returns (CAAR) over the first three trading days (days 0–2) are 0.25% (95% confidence interval 0.05–0.46%), driven almost entirely by days 0 and 1, and the cumulative effect is no longer statistically distinguishable from zero when the horizon is extended beyond about three days. Across the 20 specifications varying the abnormal-return model and estimation choices, the day-2 point estimate ranges from 0.15% to 0.5%. Second, we find no certification premium: reactions are statistically indistinguishable across SBTi-validated, SBTi-committed and unlabelled pledges. Markets appear to price the public act of commitment rather than the identity of its certifier. Third, the policy environment matters: announcement returns are statistically significant only for US-listed firms; they do not increase with the effective carbon rate faced by the firm but are larger for sectors not covered by any carbon price regulation scheme; and within the United States, the political anti-ESG backlash that began in 2022 is associated with a relative decline of about 2 percentage points in announcement returns for firms headquartered in anti-ESG states - alongside a marked decline in the US share of global commitment announcements.

To our knowledge, this is the broadest announcement-level event-study evidence on corporate climate pledges to date: press-release time stamps allow daily identification across a global, multi-sector sample, where existing announcement studies rely on US news text [24], single industries [25, 29], or the annual dating of commitment registries [11]. The broader sample changes the conclusion: the average premium we measure is smaller and more short-lived than the persistent US reactions documented by Bauer [24], and concentrated precisely in the United States. It is also, to our knowledge, the first study to test directly whether third-party certification commands an announcement premium, and to document that the market value of climate pledges is contingent on the local political economy, connecting the pricing of corporate pledges to the literature on climate policy and asset prices [27, 28, 30, 31] and on the financial costs of the US anti-ESG backlash [32].

Results

Markets react positively but modestly. Figure 1 shows the cumulative average abnormal return around the mitigation pledge announcement day for the 558 retained announcements,

where abnormal returns are computed from a market model estimated against the firm’s country benchmark index over the 75 trading days preceding the event window (Supplementary Note). The pre-event window shows no significant drift, supporting the information content of the announcement date itself. At the announcement, prices react immediately: The cumulative effect over days 0–2 is 0.25% (95% CI 0.05–0.46%). The estimate does not depend on how missing trading days are aggregated. Fig. 1 overlays the two polar aggregation rules : the balanced firm-level CAR, which keeps only firms with a complete set of daily abnormal returns over the window (0.28% at day +2, 95% CI 0.08–0.49%, $N = 515$), and the daily-AAR-cumulate rule, which uses every available firm-day observation (0.25%, $N = 529$ –551 per day); the two series are nearly indistinguishable at every horizon. The cumulative effect then drifts back: by day +5 it is roughly 0.10% and no longer statistically significant. The full market adjustment is therefore concentrated in the first two to three trading days, with no global evidence of a longer diffusion period at the daily horizon. The limited magnitude and span of the effect in a design built to maximize detectability is itself a central result: on the channel through which pledges actually reach the market, their marginal information content is deemed close to nil.

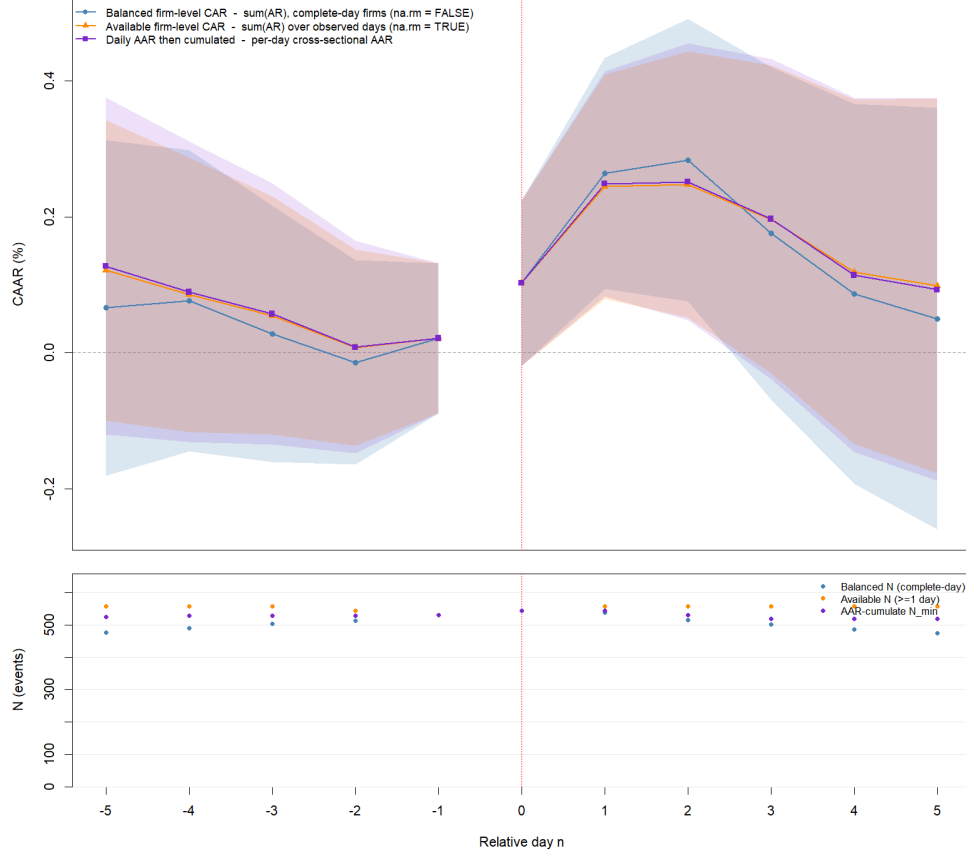


Figure 1: Cumulative average abnormal returns around commitment releases

Notes: Three aggregation rules are shown. Blue, balanced firm-level CAR: abnormal returns are cumulated within firm over the event window, retaining only events with no missing trading day; CAAR is the cross-firm mean and confidence intervals use Brown–Warner cross-sectional standard errors. Purple, AAR-cumulate: daily average abnormal returns are computed first and then cumulated over event time, with cross-period- independence standard errors. Orange, unbalanced firm-level CAR: abnormal returns are cumulated within firm using all available event-days, and CAAR is averaged across events observed at each horizon, with Brown–Warner cross-sectional standard errors. Shaded bands denote 95% confidence intervals. The lower panel reports the number of usable events per day under each rule.

Robustness. A central concern in event studies is that observed reactions reflect concomitant news rather than the commitment itself (for instance if firms bundle climate pledges with earnings news or time them around global climate events). Indeed, while there are releases all year long, and not excessively concentrated around any specific event during the timeline, the monthly distribution displays important kinks before and during recurrent annual events such as the COP meetings and the Earth Days (Supplementary Figure 1). The

0.25% day-2 estimate is robust to excluding announcements made during earnings windows, announcements bundled with other material corporate news, and announcements made during COP23–COP28 summits (both exact COP days and a ± 7 -day extended window) (Fig. 2), and balanced-rule companions of both forest panels, in which every specification is re-estimated on complete-case firm-level CARs, deliver the same pattern (Supplementary Fig. 5). Concomitant events or strategic timing around other news does not explain the premium.

Excluding any one sector group (Energy and Mining; Finance and Technology; Industry and Utilities) leaves the estimate between 0.20% and 0.29%, so no single sector drives the result, and the estimate is similar for SBTi-related announcements only (0.36%, $N = 325$) and for first or single commitments only (0.25%, $N = 462$).

Our findings are also robust to alternative modeling of abnormal returns; more precisely, to alternative benchmark indices; to alternative abnormal-return models; to various estimation windows; to inserting a buffer between the estimation and event windows; to alternative censoring rates; to dropping firms for which missing equity returns are larger than 5% or 10%; and to value-weighting in averaging abnormal returns to obtain CAARs (Extended Data Fig. 3). Across the 20 specifications varying the abnormal-return model and estimation choices, the day-2 point estimate ranges from 0.15% to 0.5%, and balanced-rule companions of both forest panels, in which every specification is re-estimated on complete-case firm-level CARs, deliver the same pattern (Supplementary Fig. 4).

The premium attaches to the disclosure, not the certifier. Fig. 3 estimates all label, region and sector contrasts jointly (Extended Data Table 1 reports the coefficients); the subgroup CAARs below are its univariate counterparts. If markets valued pledges mainly through the credibility of third-party certification, reactions should differ sharply across certification labels. They do not (Extended Data Fig. 2). Day-2 CAARs are 0.45% for SBTi-validated targets ($N = 193$), 0.25% for SBTi targets not yet validated ($N = 131$), and 0.2% for unlabelled new targets ($N = 85$); the cross-label differences are not statistically significant at conventional levels.

The OLS multivariate regression confirms the picture: no label indicator is significantly different from zero, sector differences are small (the largest, Finance and Technology relative to Industry and Utilities, is +0.3 percentage points, significant only at the 10% level). The one negative point estimate is for GFANZ-affiliated announcements (-0.6% in the subgroup figure; -0.9 percentage points, significant at the 10% level, in the multivariate specification), but this cell contains only 18 events, and the alliance later unravelled.

Target modifications — restatements or upward revisions of existing targets — elicit

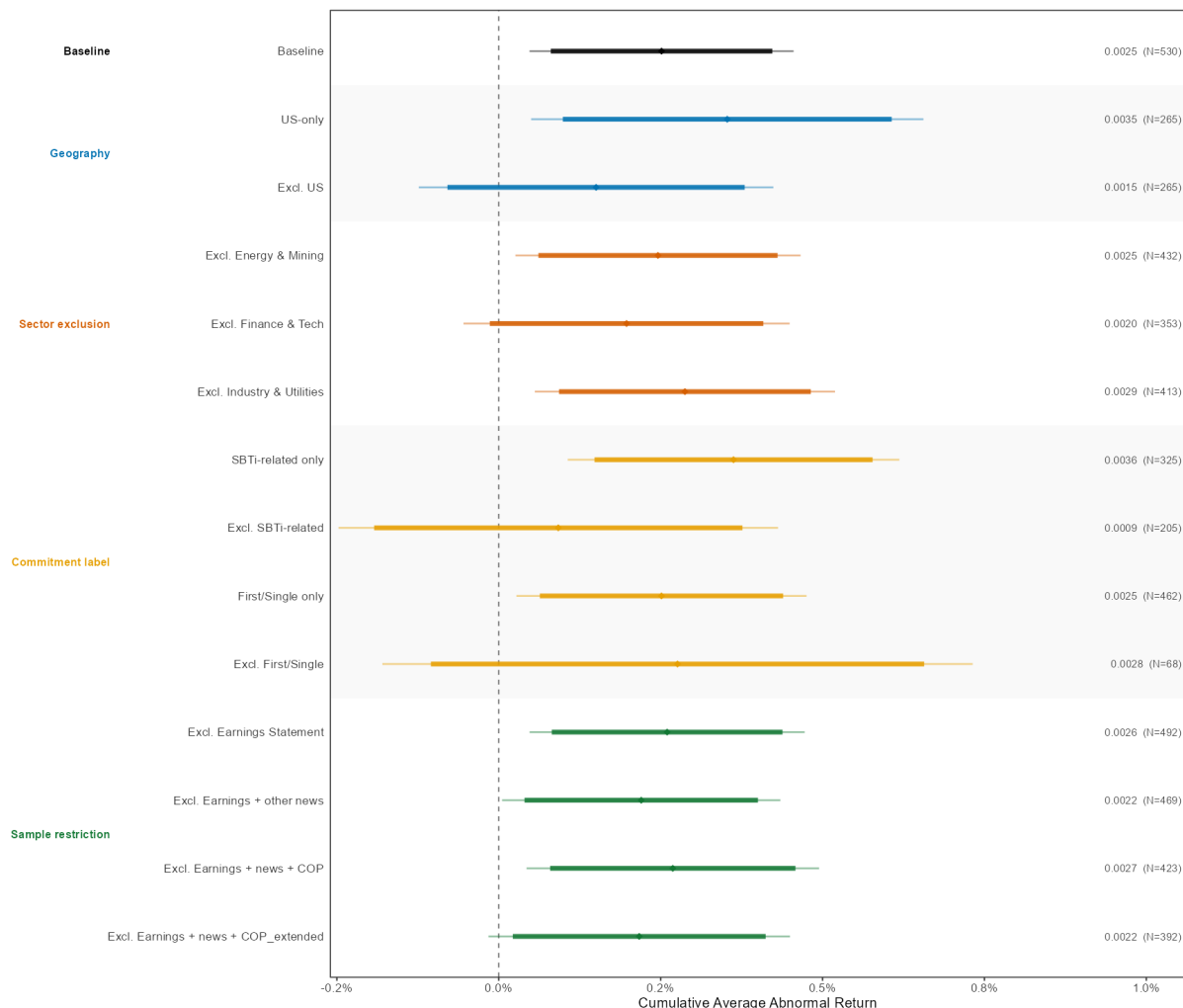


Figure 2: Robustness checks: geography, sectors, labels and sample restrictions

Note: Day-2 CAAR (AAR-cumulate rule) across additional specifications; thick and thin bars are 90% and 95% confidence intervals (cross-period-independence standard errors, $df = N_{\min} - 1$). Sample restrictions exclude, cumulatively, announcements within earnings windows, announcements bundled with other material news, and announcements during COP summits (exact days, then ± 7 days).

essentially no reaction (0.1%, $N = 101$), consistent with the market pricing the original commitment rather than its restatement.

This null is informative rather than underpowered. Focusing on target validation from the main certifier, the SBTi, the validated-minus-unlabelled contrast is 0.23 percentage points (95% CI, -0.39 to +0.86 pp). The design would detect a certification premium of 0.89 pp with 80% power, and an equivalence test rejects, at the 5% level, any premium larger than 0.75 pp (see Supplementary Note 2 for the method). Any third-party validation premium is therefore bounded at roughly three times the 0.25% average announcement effect and is statistically indistinguishable from zero.

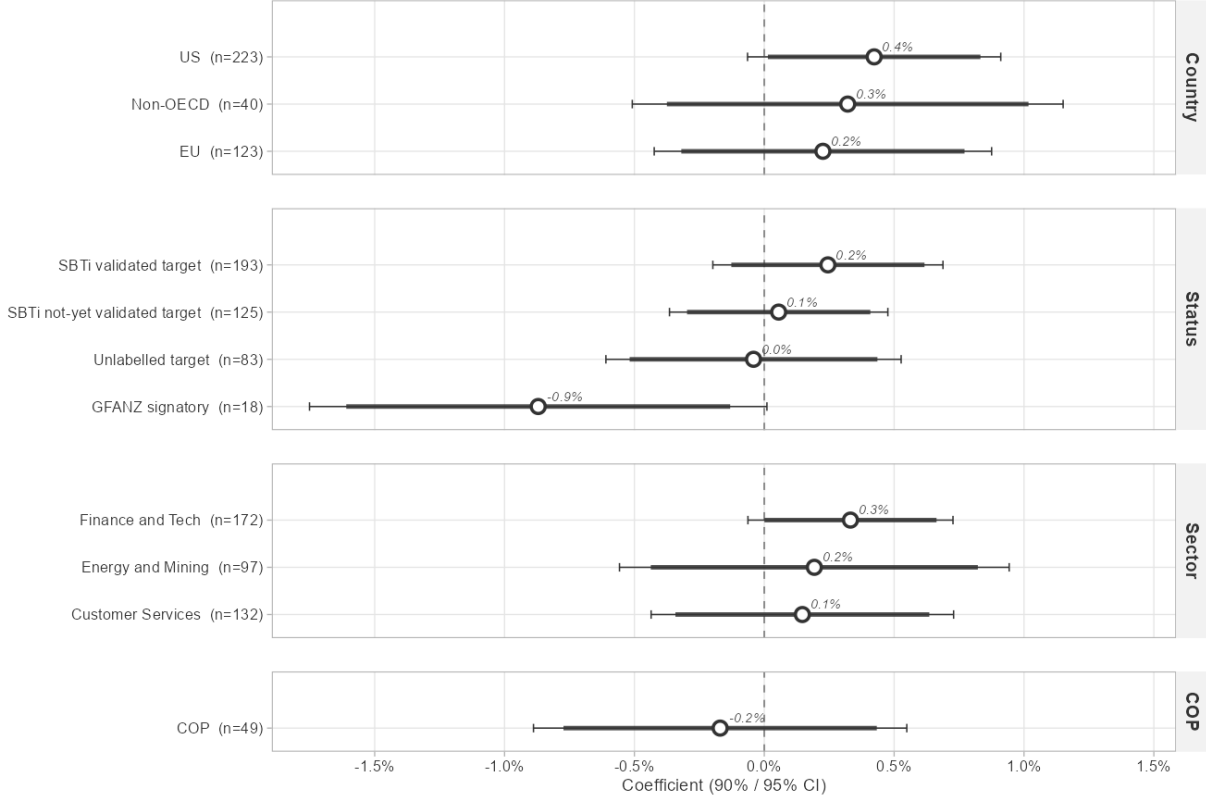


Figure 3: Multivariate analysis of day-2 CARs

Notes: OLS coefficients from a single regression of event-level $CAR(0,2)$ on region, commitment-label and sector indicators and a COP dummy ($N = 516$ complete cases). Reference categories: OECD-others (region), increased-ambition target (label), Industry and Utilities (sector). Thick and thin bars are 90% and 95% confidence intervals from CR2 cluster-robust standard errors, clustered by country of incorporation ($G = 39$). Standard errors do not propagate market-model estimation variance.

Reactions concentrate in the United States. The geographic concentration documented in the robustness panel — a significant day-2 CAAR for US-listed firms (0.35%, $N = 265$) against a small, insignificant ex-US estimate (0.15%, $N = 265$; Fig. 2) — holds across the entire event window. Extended Data Fig. 1 traces the US-only counterpart of Fig. 1: the US reaction builds over days 0–2 to 0.41% under the balanced firm-level CAR rule (95% CI 0.11–0.72%, $N = 261$) and 0.37% under the AAR-cumulate rule ($N = 265$) — about one and a half times the whole-sample estimate — with no significant pre-event drift. In the multivariate specification (Fig. 3), the US indicator is the only geographic indicator that retains significance (at the 10% level). Nor is it driven by the sectoral composition of committed US firms: excluding any one sector group leaves the whole-sample estimate between 0.20% and 0.29% (Fig. 2), and the largest sector contrast in the multivariate specification is +0.3 percentage points. The US premium is thus a genuinely geographic phenomenon — which raises the

question of what, within the US policy environment, sustains it. The within-US evidence below shows that it is politically fragile.

No carbon-price stringency premium. A natural hypothesis is that pledges are worth more where carbon pricing is stringent, because the pledge then hedges a larger regulatory cost [33, 34]. Our data do not support it. Using the OECD Effective Carbon Rates (ECR) matched to each event by country, sector and year (Methods), the day-2 CAAR by carbon-price bin is non-monotonic (Fig. 4): it is largest for announcements by firms facing no carbon pricing at all (+1.2%, $N = 31$, significant at 5%), +0.4% in the below 1 €/tCO₂ bin ($N = 203$, significant at 5%), statistically insignificant in the intermediate bins — with a negative point estimate (−0.4%) in the 25–50 € bin — and weakly positive (+0.5%, $p < 0.10$) above 50 €/tCO₂. In regressions of event-level CARs on the log effective carbon rate (with and without free-allocation of emission quotas adjustment, firm emissions and the share of committed peers; $N = 422$ complete cases), the ECR coefficient is small, negative and statistically insignificant in every specification (full estimates in the replication outputs). One interpretation could be anticipation: where carbon pricing is stringent, commitments are more predictable so the surprise component of an announcement is smaller [10, 35]. The large reaction in the unregulated bin is consistent with this reading but rests on 31 events. It also suggests that carbon disclosure can be used by investors as a weak replacement to missing carbon price regulations. To better understand the role of regulation and policy ambiguity—and given that the market response to commitments at the global level is driven largely by US firms—we examine the United States more closely in what follows.

The US anti-ESG backlash erased the premium. The sharpest evidence that the political environment conditions the value of climate disclosure comes from within the United States. From early 2022, following BlackRock’s January 2022 letter endorsing stakeholder capitalism, Republican-led states adopted anti-ESG legislation and rhetoric, while other states maintained or reinforced pro-ESG policies [32]. This divergence, within a single national market, lets us compare the market reaction to announcements by firms headquartered in anti-ESG states (red states) with the reaction for firms in pro-ESG and neutral states (blue states), before and after the backlash.

Figure 5 shows the result. Before 2022, announcement returns in the two groups are statistically indistinguishable; if anything, point estimates are higher in the states that would later adopt anti-ESG policies (+1.3% pre-2021, against −0.8% in pro-ESG states; both cells are small, $N = 13$ and 12). We are explicit about what it implies for identification: the design assumes that, absent the backlash, returns in the two state groups would have evolved in

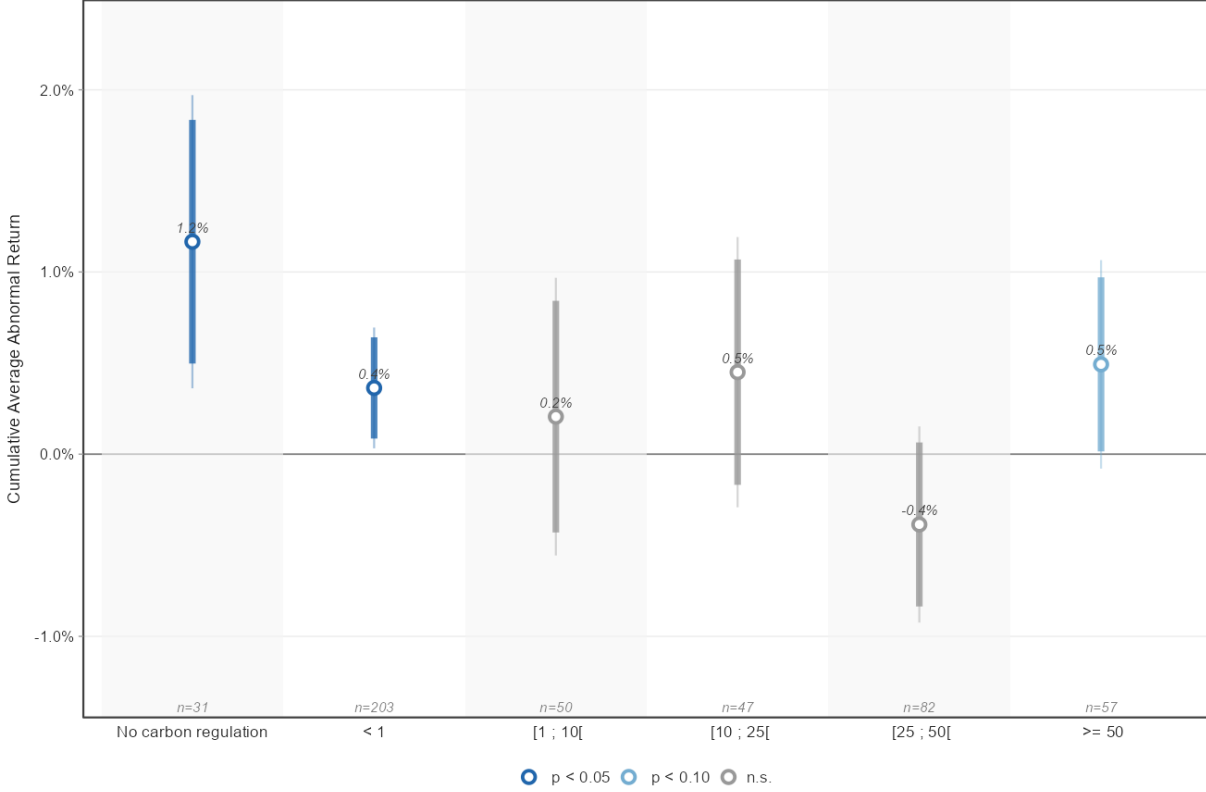


Figure 4: Day-2 CAAR by effective carbon rate

Notes: Day-2 CAAR by bin of the OECD Effective Carbon Rate including free allocations (€/tCO₂), under the AAR-cumulate rule (daily abnormal returns averaged across the available firms each day, then cumulated over the CAR(0,2) window), for the 491 of 558 events with a matched ECR. Per-bin CAAR uses cross-period-independence standard errors ($df = N_{\min} - 1$). Thick and thin bars are 90% and 95% confidence intervals. Bin sample sizes range from 31 (no carbon regulation) to 203 (<1 €). The balanced firm-level CAR companion is reported in Supplementary Fig. 6. Confidence intervals are not adjusted for multiple comparisons across bins.

parallel; the pre-2021 gap is noisy and, if anything, runs opposite to the post-2022 gap, so the assumption leans on the 2021 cells ($N = 34$ and 63), where the two groups are closely aligned, and we cannot rule out differential pre-trends on the small pre-2021 cells. After January 2022, the pattern reverses: announcement returns drop to -0.8% (not significant) in anti-ESG states while remaining positive ($+0.6\%$) in pro-ESG and neutral states. In a difference-in-differences regression on the 230 US-headquartered announcements (CAR(0,2) on a pro-ESG state indicator, a post-2022 indicator and their interaction), the interaction is $+2.0$ percentage points, rising to $+2.2$ percentage points with size, emissions, SBTi and sector controls (both significant at the 1% level with CR2 standard errors clustered by state of headquarters — the level at which the anti-ESG treatment varies, $G = 34$; Extended Data Table 3).

The interaction remains significant at the 5% level or better when we instead cluster

by country of incorporation (the much coarser $G = 7$ scheme) and at the 1% level with heteroskedasticity-robust standard errors (Supplementary Table 3). Equivalently: after the onset of the anti-ESG campaign, firms headquartered in anti-ESG states lost the announcement premium that comparable firms in other states retained.

The estimate survives the two most natural challenges to the design (Supplementary Table 4). The first concerns treatment timing: state-level anti-ESG legislation predates the Fink letter (Texas SB 13 and SB 19 were signed in June 2021), so the 1 January 2022 threshold could mistime the shock. Re-defining the post period from 1 June 2021 halves the interaction to +1.1 percentage points and renders it insignificant, and the specification loses essentially all explanatory power ($R^2 = 0.016$ against 0.048 at the baseline threshold). Far from undermining the design, this timing sensitivity is informative: had the first state statutes driven the repricing, the mid-2021 threshold should recover an effect at least as strong. That it instead attenuates the contrast — by mixing seven months of pre-backlash announcements into the treated period — indicates that markets repriced climate disclosure only once the backlash escalated into a coordinated national campaign in early 2022, not upon the first legislative signals. The second challenge concerns classification: Virginia and North Carolina, both governed by Democrats when the key legislation passed, are the most debatable entries in the anti-ESG group. Excluding them leaves the interaction unchanged at +2.2 percentage points, significant at the 1% level.

The backlash shows up in disclosure behaviour as well as in prices. The US share of global commitment announcements in our sample falls from 54% in 2021 to 42% in 2022 and 31% in 2023 (Fig. 5, bottom panel), with the steepest decline among firms headquartered in anti-ESG states. This pattern is consistent with media accounts of “greenhushing” — firms muting or concealing environmental commitments to avoid political exposure [36], although our data cannot separate fewer commitments from fewer public releases.

Discussion

Markets price corporate climate pledges as modest, short-lived information events. The average announcement raises firm value by about a quarter of a percentage point within two to three trading days, economically small relative to the capital expenditures that deep decarbonization requires, but a real and robust premium on the act of public commitment. We read this premium as a lower bound on detectability rather than a noisy estimate of a larger latent effect, and this is our central result. Our sample is selected precisely on market-facing salience: we retain only pledges that reached investors through the LSEG financial-press wire — a feed monitored by more than 500,000 finance professionals — and

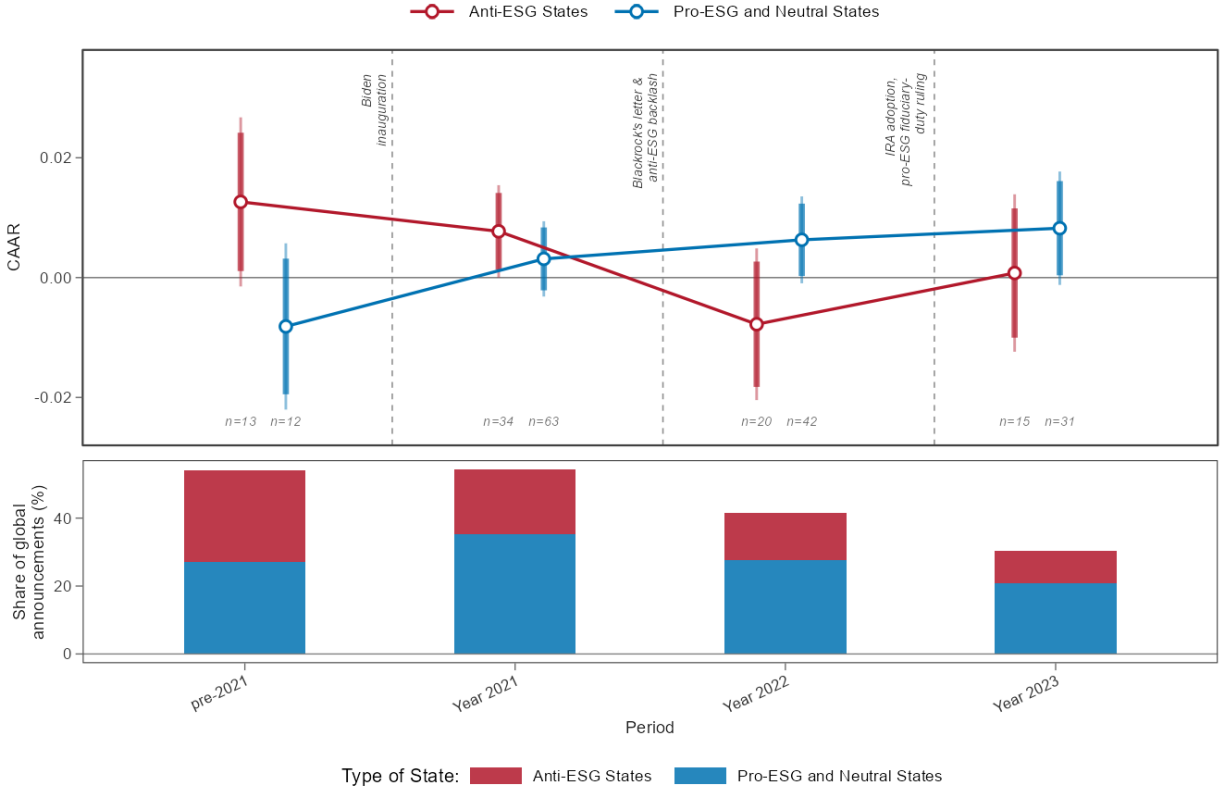


Figure 5: The US anti-ESG backlash and the market response to commitment announcements

Notes: Top: day-2 CAAR for announcements by firms headquartered in anti-ESG states (red) versus pro-ESG and neutral states (blue), by period (pre-2021, 2021, 2022, 2023); cell sizes shown at the bottom of the panel. Thick and thin bars are 90% and 95% confidence intervals (cross-period-independence standard errors). Vertical lines mark the Biden inauguration, the January 2022 BlackRock letter and onset of the anti-ESG backlash, and the IRA adoption. Bottom: share of global announcements accounted for by US firms, split by state group. State classification is described in the Methods.

we date each to the day it became public. Any attenuation from information failing to reach investors, or from imprecise timing smearing the abnormal return across days, therefore works against us. That a premium this small and this short-lived is what survives a design built to maximize detectability is informative in itself: on the channel through which pledges actually reach the market, their marginal information content is close to nil.

What this design cannot by itself separate is low valuation from full anticipation. Under market efficiency only the surprise component of a pledge moves prices, so a commitment the market already expected is, efficiently, non-news. For the question that motivates the paper, however, the two readings carry the same implication: going public with a climate target adds little to firm value at the margin, whether because investors discount the content or because they had already priced it. Either way, the market-based case for voluntary corporate decarbonization is weaker than the “it pays to be green” narrative typically presumes.

Three features of this premium discipline its interpretation.

First, the premium attaches to the disclosure, not to the certifier. Reactions are statistically indistinguishable across SBTi-validated, SBTi-committed and unlabelled pledges, and restatements of existing targets elicit no reaction. In a context of high information asymmetry, investors appear to treat the public, dated act of commitment as the informative signal, with the identity of the validator playing at most a modest role at the announcement stage. This need not mean certification is worthless: certified targets may matter for emissions outcomes [10, 11], for lending relationships [37, 38], for inclusion in net-zero-aligned portfolios [39], or at validation milestones that can fail. But it does mean the announcement-day premium cannot be read as a price on third-party verification. Evidence from the banking sector points the same way: as the Net-Zero Banking Alliance rose and then unraveled, only founding members' joining announcements moved prices, while later signings and the wave of exits left markets unmoved [40]. The asymmetry between announcement and delivery sharpens this reading. Examining the cohort of corporate emissions targets that came due in 2020, Jiang et al. [17] find that almost one in ten failed and nearly a third quietly disappeared, yet the failures attracted almost no media coverage and no detectable market reaction, whereas the initial announcements of those same targets had been rewarded with improved media sentiment and environmental scores. Ex-post accountability for pledge outcomes is thus weak: investors reward the public act of commitment at a moment when its fulfillment is largely unverifiable, which is consistent with the premium we document attaching to the disclosure itself rather than to expected delivery.

Second, the premium is geographically and politically contingent. It is statistically significant only in the United States; it does not rise with carbon-price stringency; and within the United States it disappeared entirely in anti-ESG states from 2022 onward. The carbon-price pattern, with the largest reactions where carbon pricing is absent, is consistent with an anticipation mechanism — where regulation makes commitments predictable, announcements carry little surprise [35] — and, more tentatively, with pledges serving as substitutes for regulation in unregulated environments rather than complements to it. We stress that this is an interpretation the cross-sectional evidence permits, not one it establishes: the relevant samples are small, effective carbon rates co-move with many institutional factors, and the binned contrasts are unadjusted for multiple testing.

Third, the within-US evidence points to a political-economy channel that fragments the value of climate transparency across jurisdictions. After the anti-ESG backlash, the announcement premium in exposed states was erased — a roughly 2-percentage-point relative decline — and US firms' share of global announcements fell by more than twenty percentage points, a pattern consistent with greenhushing. For firms, this turns the disclosure decision

itself into a political calculation; for policymakers, it suggests that hostility to ESG finance does not merely reallocate returns but can mute corporate transparency about transition plans.

The likelihood-to-commit (publicly) margin reinforces the anticipation reading of the cross-country results: firms most exposed to carbon pricing are more likely to commit [10], so in stringent-regulation environments commitments are both more common and more expected.

Limitations. Several caveats bound these conclusions. Our estimand is the value of going public with a pledge that the firm chose to broadcast on the financial wire, which shapes interpretation in two opposite directions. On one side, sampling on publicized, market-facing releases is what licenses the lower-bound reading above: it removes the concern that a near-zero reaction reflects information never reaching investors. On the other side, firms plausibly broadcast the disclosures they expect to be received favourably, so the measured premium is, if anything, an upper bound on the average reaction across all pledges — which reinforces rather than weakens the conclusion that the marginal pledge moves prices little. We also cannot, within this design, distinguish low valuation from full anticipation; we are explicit that our headline claim concerns the marginal information content of pledges on this channel, not investors’ ultimate valuation of decarbonization.

The remaining caveats are familiar to the design. (i) *Anti-ESG design*. Headquarters location proxies exposure to state-level pressure imperfectly (firms operate across states), so the DiD estimate is plausibly a lower bound on the true differential, and pre-2021 state-group differences — noisy but opposite-signed — mean the parallel-trends assumption leans on the 2021 period. The post date (1 January 2022) approximates the Fink letter. (ii) *Selection and coverage*. The LSEG press aggregator over-represents large firms and advanced economies, and our English-keyword search adds a language bias; our estimates speak to large listed firms that publicize their commitments. Within that population, the patterns of who commits (large firms, emission-intensive sectors, Europe for the SBTi) is extremely consistent with the main findings documented for commitments at large [10, 24]. (iii) *Outcomes*. We measure market valuation, not abatement: nothing here speaks to whether pledges are kept [10, 11], and the analysis is confined to equity markets, whereas transition risk is also priced in credit [38].

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Methods

Commitment announcements from press releases

We extract press releases from the LSEG (ex-Refinitiv) press aggregator up to December 31st, 2023. LSEG started recording companies commitments in 2017 and is generally considered as one of the most reliable and exhaustive gatherers of financial news. This database aims at aggregating all press releases and financial media coverage of those releases. This technique of collection allows us to access the precise date of release to market. Each release also carries an identifier associated with the company’s main equity (the Refinitiv Identifier Code, or RIC), hence limiting the risk of error within the matching process.

The LSEG Press database relies on more than 5,000 global press sources; it allows us to get a broad view on the broadcast data. Furthermore, it has the advantage of being a media outlet of reference, making sure the information does indeed make it through towards effective investors.

We first handle this search by looking, within the press release space, to all releases displaying one of a group of keywords in their headlines,—“Climate Pledge”, “science-based”, “science based”, “SBTi”, “carbon target”, “net-zero”, “Net Zero”, “Climate Commitments”, “emissions target”, “emission target”, “emissions reduction target”, “emissions reduction goal” and “Environmental Commitment”—,or categorized by LSEG as related to “carbon neutrality”. We obtain 947 press releases announced between 2017 and 2023. A typical new net zero commitment typically includes standard wording for the title, such as in Supplementary Figure 2; as such the margin of error in the categorization is quite limited.

We recorded other kinds of information relative to the target, such as the year of the interim target (when any), the percentage of reduction, the year of the net zero target. We also recorded whether the commitment was part of a coalition of peers announcements (in which case the observation was dropped from the sample) and whether the press release contained any information that was not related to the commitment (in Extended Data Figure 3, we check that dropping these observations does not affect the point estimate).

From each release, we extract the date of announcement, whether it is a new net zero commitment. We classified as “New” commitment releases related to new net zero target or related to a particular step of the SBTi process (commitment, submission, or validation), the interim target (consisting in a target year, an objective for scope 1&2 emissions and a base year), the date at which the firm plans to be net-zero for its scope 1&2 emissions. When several interim targets are mentioned, we noted the most stringent one, discounting time linearly. We acknowledge that linear discounting might present an issue, since firms which have set a target on a longer time-horizon were shown to respect less their objectives [9].

However, there is no clear evidence in support for any better discounting process. We also account for the label that validated this target if any; we focused on the following labels: The Science-based Target initiative (SBTi) and the Glasgow Financial Alliance for Net Zero (GFANZ), that groups the Net-Zero Banking Alliance, the Net-Zero Insurance Alliance and the Net-Zero Asset Managers. Furthermore, we considered the company to be in a SBTi process if it explicitly stated it (while some companies simply mention SBTi without actually stating they are part of the process). We further classified the progress within the SBTi process depending on whether the company simply sent a letter of commitment, set new goals consistent with the SBTi standards, or was validated by the SBTi. For example, in Supplementary Figure 2, the company announces its validation by the SBTi.

Second, we checked on each company website the exact date of release of the statement. Indeed, we noticed that there could exist a lag of several days between the official release and the recording by the Refinitiv database, which would undermine a sharp short event study. Supplementary Figure 3 shows the distribution of these gaps. We only kept those releases that we could clearly match to a release date (either because the press release itself was recorded on the company website or because some financial media mentioned the exact date of the information release by the company). Furthermore, when we found any additional relevant release from the same company, we added it to the sample.

Day 0 convention. If an announcement occurs on a non-trading day, Day 0 is defined as the first subsequent trading day. Specifically, the algorithm searches for the next available trading date within a maximum window of five calendar days (Day +0 to Day+5). This convention follows a standard practice in event study methodology.

Analysis samples

The main event-study sample comprises 558 press releases emitted by listed firms between 2017 and 2023 and retained after the standard event-study data-preparation pipeline (verification of release dates, exclusion of firms whose country of exchange falls outside the LSEG FR Global Equity universe, removal of events for which the local equity series is flat over the post-event horizon, and removal of events for which insufficient pre-event return data is available). Sample composition by country, sector, year and commitment label is reported in Supplementary Table 1. Index returns are complete over every event’s estimation window. Stock returns are not: 90% of retained events have at least one missing stock return in the 75-trading-day estimation window (maximum missing share 14.5%, below the 15% baseline gate), and the stricter 5% and 10% missingness gates used as robustness checks are binding (they retain 160 and 441 of the 558 events, respectively; see the robustness forest).

Some robustness checks and the heterogeneity analysis involve additional sample restrictions due to missing values in the covariates: the multivariate analysis is run on 516 events with a non-missing complete-case CAR(0,2) and label classification; 491 events have a matched OECD Effective Carbon Rate (the binned carbon-price figure uses these), and the carbon-pricing regressions are run on the 422 events that in addition have non-missing emissions and peer-share covariates (shared complete-case mask across all specifications; estimates in the replication outputs).

Firm characteristics and matching

LSEG Monitor App data universe is composed of the 67,900 active listed firms. This database is used for all firms time-invariant or yearly characteristics (when the variable of interest is year-variant, we retrieve the data from the year of the commitment). One important stake with this study was to precisely understand the nature of each variable on LSEG, since there are often multiple variables with very close definitions. We extracted and used the following variables (in brackets the LSEG function associated to each variable): the identifier i.e. the RIC of the main equity, the ISIN Code (TR.Code), the company name (TR.CompanyName), the company location in terms of Country of Exchange (TR.ExchangeCountry), Country of Incorporation (TR.RegistrationCountry) and Province or State of Incorporation (TR.RegStateProvince) and Province or State of Headquarters (TR.HQStateProvince), the main sector of activity following the TRBC classification in economic sectors (TR.TRBCEconomicSector) and in industries (TR.TRBCIndustry), the Company Market Capitalization (TR.CompanyMarketCap) (we extracted the Company Market Cap and not the Main Equity Market Cap, because the latter variable is not standardized across firms), CO_2 emissions in scope 1 (TR.CO2DirectScope1) and in scope 2 (TR.CO2IndirectScope2).

We match the event data with the LSEG financial data collected through the *Screener App*. The matching process relies on the use of the Refinitiv Identifier Code (RIC) for the PR sample. The relevant RIC is in general associated with the news release. When it is not, or when the RIC was not associated with the ordinary share of the firm, we completed this work manually. The RIC is an identifier at the equity and market level. However, in *LSEG Screener App*, each listed firm is associated to a unique RIC of reference, that happens to be the RIC of their primary securities, while excluding dual listing. It allows us to rule out any risk of double counting or mismatching.

Share Committed. *Share Committed* denotes the proportion of LSEG-covered firms having made a net-zero or science-based climate commitment within a given country–sector–year

cell, and serves as a proxy for the degree of climate maturity within a firm’s competitive environment.

Stock and Index Returns

We extract from LSEG Datastream the daily close prices (TR.ClosePrice) for all firms within our sample between January 1st, 2014, and January 31st, 2024. The close price is the last trade price of the company main equity in a given day. Similarly, we extract from Datastream the daily close price of each country index of reference in order to compute the market model.

For the market return, we use the country-based FR Equity Price Return Indices for all countries in the baseline, and we alternatively use the S&P 500 as a robustness. The FR Equity Price Return Indices is a family of free-float-adjusted market-capitalization-weighted indices that are comparable amongst them because they are made up following the same methodology. LSEG compute them for 51 developed or emerging countries; as a consequence, we drop certain datapoints that were drawn from countries that do not belong to LSEG Global Indices universe (three commitments from Romania, Slovenia, Kenya.) or for which we only had one observation (three commitments from Peru, Iceland and Colombia). All further methodological details are provided in [42].

For each country, LSEG emits two indices: a Price Return Index, that is based on the overall free float market capitalisation of the constituents, and a Total Return Index, that accounts for any gross dividend payment that takes place. As it is standard in the short-event study literature, we used the Price Return Indices.

Effective Carbon Rates Data

Effective carbon rates are drawn from the Detailed Effective Carbon Rates database of the [43], which provides economy-wide estimates of the price on carbon emissions resulting from fuel excise taxes, carbon taxes, and emissions trading systems. The key variable used in the regressions is the *Effective Carbon Rate including Free Allocations* (`ecr_incl_fa`), which adjusts the gross ECR downward to account for the implicit subsidy granted to firms receiving free allowances under cap-and-trade schemes. This variable (*Discounting Free Allocations*) differs from the unadjusted ECR for 93.5% of matched observations. Using the free-allocation-adjusted measure is the more conservative choice, as it reflects the actual carbon cost borne by regulated firms. Separately, 31 matched observations carry an ECR (without free allocation) of exactly zero; these correspond to firms in countries or sectors where no carbon pricing instrument is in place, and are retained in the analysis under the “No price” category.

The matching of companies with appropriate carbon prices is detailed in the Supplementary Note 3.

Multivariate specification

For the multivariate analysis, we compute each event’s CAR(0,2) as in Eq. S4 and regress it by OLS on a set of category dummies:

$$CAR_i(0, 2) = \alpha + \mathbf{Country}_i' \beta + \mathbf{Label}_i' \gamma + \mathbf{Sector}_i' \delta + \theta COP_i + \varepsilon_i, \quad (1)$$

where $\mathbf{Country}_i$ is a vector of region dummies (EU, US, non-OECD; reference: other OECD countries), $\mathbf{Label}_i$ a vector of commitment-label dummies (GFANZ signatory, SBTi not-yet-validated target, SBTi validated target, unlabelled target; reference: increased-ambition target), $\mathbf{Sector}_i$ a vector of sector dummies (Customer Services, Energy and Mining, Finance and Tech; reference: Industry and Utilities), and COP_i indicates an announcement released during a COP window. For Sectors, we group the 11 sectors from the LSEG TRBC Classification into 4 groups (Energy and Materials are grouped into “Energy & Mining”; Industrials and Utilities into “Industry & Utilities”; Consumer Cyclical, Consumer Non-Cyclical and Healthcare into “Customer Services”; Real Estate, Financials and Information Technology into “Finance & Tech”). Standard errors are CR2 cluster-robust, clustered on the country of incorporation. The estimates are reported in Extended Data Table 1.

Anti-ESG difference-in-differences

We first compute average CAARs for each of four periods (pre-2021, 2021, 2022, and 2023) and for two groups of US states—anti-ESG states, and pro-ESG or neutral states. These descriptive cell-level estimates are plotted in main-text Figure 5 and reported in Table 2.

We then use a cross-sectional difference-in-differences. Unlike the rest of the analysis, we do not use the state of incorporation, because companies choose their state of registration in a manner that often is unrelated to the place of their actual economic activities, most of the time choosing Delaware for fiscal reasons. Hence, using the state of registration would have made this study ineffective. On the contrary, the state of headquarters is a convenient tool, since the decision is often made to locate the headquarters in a state that is meaningful and consistent with the company economic activity. Yet the matching is not exact, and does not take into account the potential diversity in the states where the company operates. Therefore, one should remain careful that the estimate we measure here is a lower bound of

the real effect of the differential impact of state regulation of the returns on the company environmental commitments. Our specification estimates how the performance of companies located in pro-ESG or neutral states is affected by net-zero commitments during the ESG controversy as compared with companies headquartered in anti-ESG states:

$$\begin{aligned} CAR_i = & \alpha + \beta_1 \times \mathbb{1}\{t \geq t_0\} + \beta_2 \text{BlueState}_i \\ & + \beta_3 \text{BlueState}_i \times \mathbb{1}\{t \geq t_0\} + \mathbf{X}_i \eta + \varepsilon_i \end{aligned} \quad (2)$$

where $\mathbb{1}\{t \geq t_0\}$ and BlueState_i are respectively time- and space-varying binary variables. The threshold t_0 is fixed at January 1, 2022, which approximates the release date of Larry Fink’s January 2022 letter to CEOs and matches the period split used in Figure 5; the 2021 period begins at Biden’s inauguration (20 January 2021); releases from 1–19 January 2021 are assigned to the pre-2021 bin. BlueState_i equals one if the company’s headquarters are located in a pro-ESG or neutral state. We estimate Equation 2 on the subsample of commitment releases for which the related company has its headquarters within the United States ($N = 230$ events with a state classification: 82 in anti-ESG states, 148 in pro-ESG or neutral states; descriptive statistics by state group are reported in Table 2).

State classification. We classify as *anti-ESG* (Red states): Alabama, Alaska, Arkansas, Florida, Georgia, Idaho, Indiana, Iowa, Kansas, Kentucky, Louisiana, Mississippi, Missouri, Montana, Nebraska, North Carolina, North Dakota, Ohio, Oklahoma, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, West Virginia and Wyoming; and as *pro-ESG or neutral* (Blue states): Arizona, California, Colorado, Connecticut, Delaware, Hawaii, Illinois, Maine, Maryland, Massachusetts, Michigan, Minnesota, Nevada, New Hampshire, New Jersey, New Mexico, New York, Oregon, Pennsylvania, Rhode Island, Vermont, Washington, Washington DC and Wisconsin.

We classify as *anti-ESG* states all states that either (i) signed the letter asking Congress to overturn the Department of Labor’s ESG investing rule, (ii) participated in Governor Ron DeSantis’ anti-ESG alliance, or (iii) whose attorneys general were leading an investigation or lawsuit against parties engaged in ESG-related initiatives as of the end of 2023 (Texas being the sole addition from the third criterion). All other states are counted as pro-ESG states.

Key dates. The vertical markers in Figure 5 correspond to four events that structure the US ESG controversy, two of which play a direct role in the estimation. First, President Biden’s inauguration on 20 January 2021 marks the beginning of a federal regulatory environment more receptive to ESG-oriented finance; it therefore defines the boundary between the pre-2021 and

2021 periods. Second, BlackRock’s January 2022 annual letter to CEOs became a focal point for conservative opposition to ESG, particularly because ESG initiatives were often framed by Republican critics as advancing progressive priorities such as climate change mitigation and diversity [44]. The letter helped crystallize the subsequent Republican-led backlash, including state anti-ESG statutes, treasury divestment lists, and attorney-general investigations. The DiD threshold t_0 (1 January 2022) approximates this date and separates the 2021 and 2022 periods. The remaining two markers are contextual and enter no estimation: the adoption of the Inflation Reduction Act (16 August 2022), the largest federal climate-spending program to date, which sharpened the partisan salience of corporate climate positioning; and the March 2023 presidential veto preserving the Department of Labor rule allowing ESG considerations in retirement-plan investing — the first veto of the Biden presidency — which entrenched the federal–state divide over ESG finance. Anti-ESG legislation at the state level predates the backlash’s national escalation (Texas SB 13 and SB 19 were signed in June 2021) [32]; Extended Data Table 4 shows that re-dating the treatment to June 2021 attenuates the interaction, consistent with markets repricing climate disclosure only once the campaign became national in early 2022.

Controls and inference. X_i is a set of firm-level and commitment-level controls, including the type of commitment, the log market capitalisation, sector dummies and CO₂ emissions (we cannot use company fixed effects, since the DiD is cross-sectional). Standard errors are CR2 cluster-robust, clustered by *state of headquarters* — the level at which the anti-ESG treatment varies ($G = 34$ states in the baseline specification; $G = 33$ in the specifications that lose observations to missing emissions data) — and inference uses cluster-robust t-statistics with $df = G - 1$. Table 3 reports the main results and Table 3 reports an inference sensitivity analysis: the interaction coefficient remains significant at the 5% level or better under the coarser country-of-incorporation clustering used in earlier drafts ($G = 7$ very unbalanced clusters, for which CR2 inference is fragile) and at the 1% level under heteroskedasticity-robust (HC1) standard errors.

Reporting summary Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The press-release event data were collected from the LSEG (formerly Refinitiv) press aggregator, and firm characteristics and daily prices from LSEG Screener and Datastream, under licence; these raw data are available from LSEG but restrictions apply. Effective carbon rates

are from the OECD Detailed Effective Carbon Rates database (2023 edition); these data are available from the OECD but restrictions apply.

Code availability

All analyses were conducted in R version 4.3.3. The complete replication code (event-panel construction, abnormal-return estimation, robustness checks and regression tables) is available at [Zenodo DOI / GitHub URL].

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Author contributions

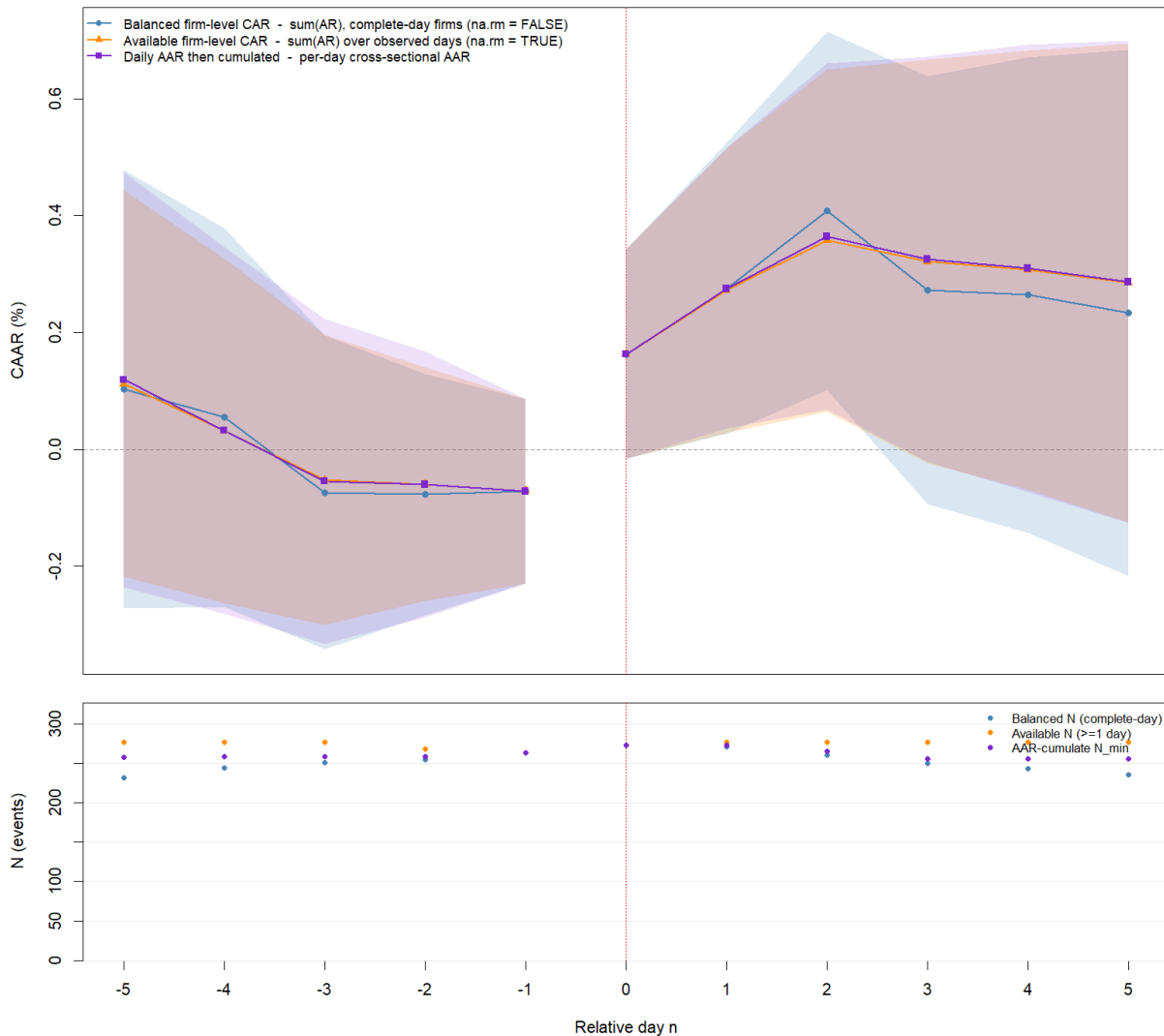
M.A.: Data curation, Methodology, Formal analysis, Writing – original draft. R.C.: Conceptualization, Methodology, Formal analysis, Writing – Review Editing, Supervision.

Both authors contributed to the interpretation of the results, critically revised the manuscript, and approved the final version for publication.

Competing interests

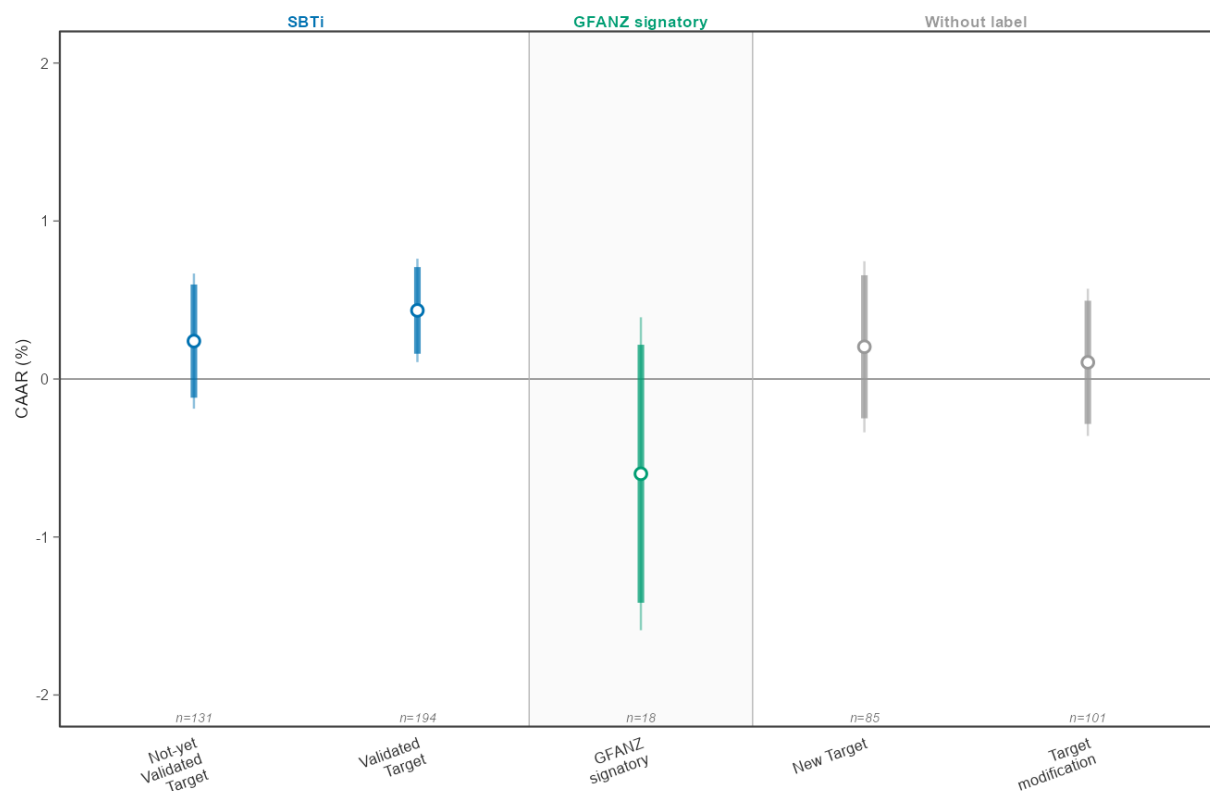
The authors declare no competing interests.

Extended Data



Extended Data Fig. 1: Cumulative average abnormal returns around commitment releases, US-listed firms only: balanced firm-level CAR versus AAR-cumulate

Notes: Same construction as the headline butterfly figure in the main text, restricted to announcements by US-listed firms (country of exchange = United States; 281 events enter the AR estimation, 277 are retained). Daily abnormal returns are the same market-model residuals as in the full sample (winsorised at 5% per tail per event day within the US subsample); only the cross-sectional aggregation is restricted to the US subsample. Blue: balanced firm-level CAR (complete-day firms; Brown–Warner cross-sectional standard errors); at day +2 the CAAR is 0.41% ($N = 261$). Purple: daily AARs cumulated over the window (cross-period-independence standard errors); at day +2 the CAAR is 0.37% ($N = 265$). Shaded bands are 95% confidence intervals; the lower panel reports the number of usable events per day under each rule. Neither interval propagates market-model estimation variance.



Extended Data Fig. 2: Heterogeneity by certification label

Notes: Day-2 CAAR by commitment type (AAR-cumulate rule; thick and thin bars are 90% and 95% confidence intervals, cross-period-independence standard errors). SBTi not-yet-validated pools the SBTi commitment and submission steps (N = 131); SBTi validated targets, N = 193; GFANZ signatories (NZIA, NZAM, NZBA), N = 18; unlabelled new targets, N = 85; modifications of existing targets, N = 101. Cross-label differences are not statistically significant at conventional levels; the GFANZ cell is small and its negative point estimate should be read with caution. Confidence intervals are not adjusted for multiple comparisons.

Extended Data Table 1: Multivariate analysis: Alternative standard errors clustering

	<i>Dependent variable:</i>				
	CAR(0,2)				
	Inc_Country	Exc_Country	Country4-Sec cell	Inc-Sec cell	Exc-Sec cell
EU	0.002 (0.003)	0.002 (0.003)	0.002 (0.003)	0.002 (0.003)	0.002 (0.003)
Non-OECD	0.003 (0.004)	0.003 (0.004)	0.003 (0.004)	0.003 (0.004)	0.003 (0.005)
US	0.004* (0.002)	0.004* (0.002)	0.004* (0.003)	0.004* (0.003)	0.004* (0.002)
GFANZ signatory	−0.009* (0.004)	−0.009* (0.005)	−0.009*** (0.003)	−0.009* (0.005)	−0.009 (0.006)
SBTi not-yet validated target	0.001 (0.002)	0.001 (0.003)	0.001 (0.002)	0.001 (0.003)	0.001 (0.003)
SBTi validated target	0.002 (0.002)	0.002 (0.003)	0.002 (0.003)	0.002 (0.003)	0.002 (0.003)
Unlabelled target	−0.0004 (0.003)	−0.0004 (0.003)	−0.0004 (0.004)	−0.0004 (0.004)	−0.0004 (0.003)
Customer Services	0.001 (0.003)	0.001 (0.004)	0.001 (0.002)	0.001 (0.003)	0.001 (0.004)
Energy and Mining	0.002 (0.004)	0.002 (0.007)	0.002 (0.003)	0.002 (0.004)	0.002 (0.005)
Finance and Tech	0.003* (0.002)	0.003 (0.003)	0.003** (0.001)	0.003 (0.003)	0.003 (0.004)
COP	−0.002 (0.004)	−0.002 (0.004)	−0.002 (0.003)	−0.002 (0.003)	−0.002 (0.004)
Constant	−0.002 (0.004)	−0.002 (0.005)	−0.002 (0.003)	−0.002 (0.004)	−0.002 (0.005)
Clusters G	39	31	16	85	72
Observations	516	516	516	516	516
R ²	0.015	0.015	0.015	0.015	0.015

Note: All five columns are the same OLS regression (Equation 1); the dependent variable is event-level CAR(0,2), computed as a complete-case sum of daily abnormal returns over $AR_0, \dots, AR_2$. Only the cluster definition varies. Col 1 = main paper specification: clustering on country of incorporation (raw, G=39). Col 2 = country of exchange listing (raw, G=31). Cols 3-5 = cell-clustering on country $\times$ Sector interaction (col 3 uses the 4-bin macro-region for the country, cols 4-5 use raw country). All five are one-way CR2 on the interaction. Reported standard errors are computed at the regression stage, using event-level CARs as outcomes. They do not include the additional uncertainty from the first-stage estimation of daily abnormal returns.

Extended Data Table 2: Anti-ESG heterogeneity: point estimates behind Figure 5

Period	ESG-state bucket	N	CAAR (%)	SE (%)	90% CI (%)	95% CI (%)	Share (%)
pre-2021	Anti-ESG States	13	1.26	0.65	[0.11, 2.42]	[-0.15, 2.67]	27.1
pre-2021	Pro-ESG and Neutral States	12	-0.82	0.63	[-1.95, 0.32]	[-2.20, 0.57]	27.1
Year 2021	Anti-ESG States	34	0.77	0.38	[0.13, 1.41]	[0.00, 1.54]	19.0
Year 2021	Pro-ESG and Neutral States	63	0.31	0.31	[-0.21, 0.84]	[-0.32, 0.94]	35.3
Year 2022	Anti-ESG States	20	-0.78	0.60	[-1.82, 0.27]	[-2.04, 0.49]	13.8
Year 2022	Pro-ESG and Neutral States	42	0.63	0.36	[0.03, 1.23]	[-0.09, 1.36]	27.7
Year 2023	Anti-ESG States	15	0.08	0.61	[-1.00, 1.16]	[-1.24, 1.39]	9.8
Year 2023	Pro-ESG and Neutral States	31	0.82	0.46	[0.04, 1.61]	[-0.12, 1.77]	20.7

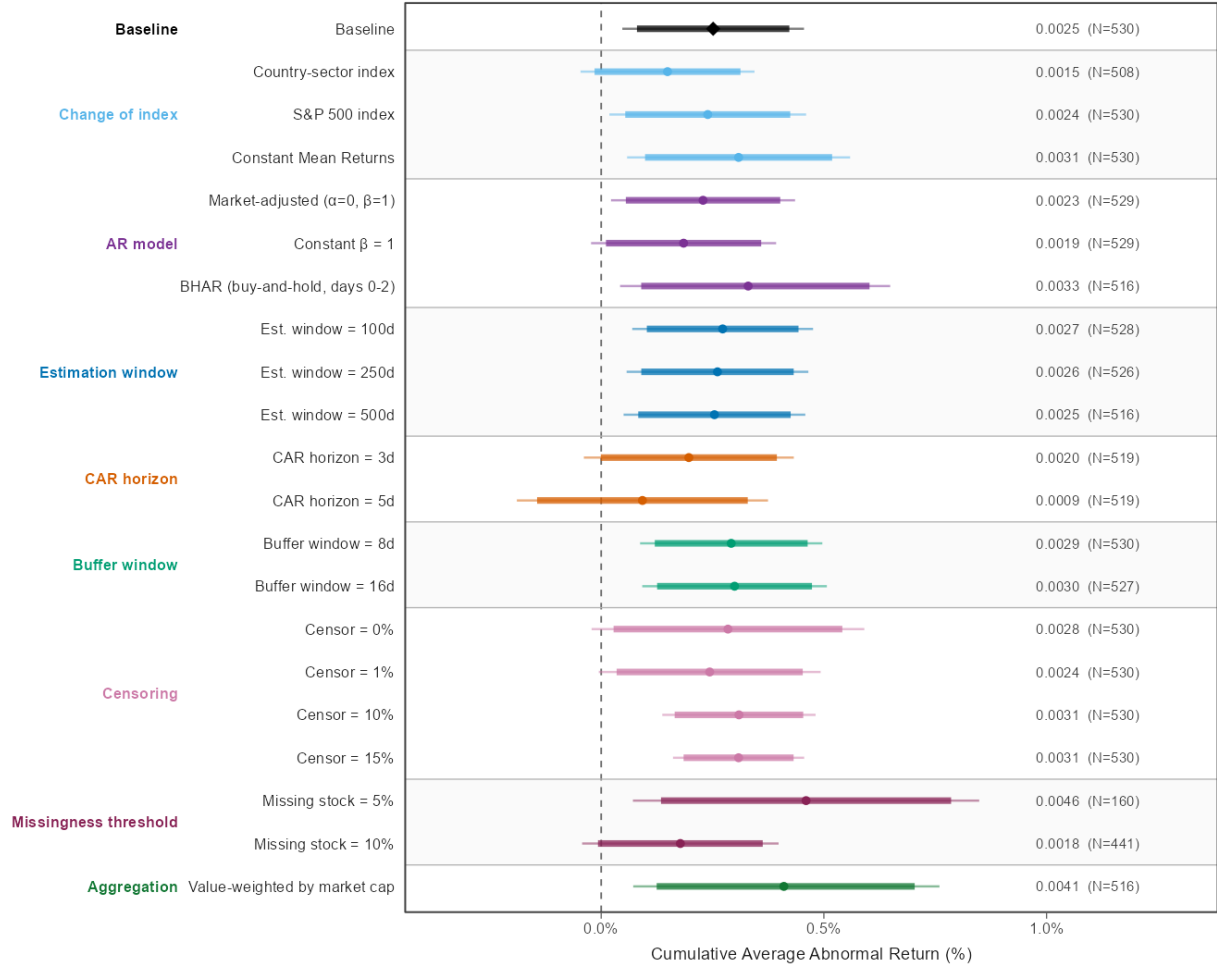
Note: Sample restricted to events whose firm is headquartered in a US state with a classified ESG stance. SE uses the cross-period-independence rule and Student-t critical values with $N_{min} - 1$ df, where N_{min} is the smallest daily cell count over the cumulation window. Share = the cell's fraction of *all* commitment announcements in the period (US + non-US), so Pro-ESG + Anti-ESG shares sum to the US share of the period.

N convention: N is the minimum daily cross-sectional count over the CAR(0, 2) window (AAR-cumulate rule). The DiD regression (Extended Data Table 3) uses complete-case CAR(0, 2), so its N may be 1–2 events smaller. Share values use raw period totals computed before the CAR availability filter; identical share values in two cells of the same period reflect equal raw group counts. Analysis period: 2018-01-01 (period start) through end of sample; 2017 events are excluded.

Extended Data Table 3: Anti-ESG DiD on US-headquartered commitment releases

	CAR(0,2)			
	(1)	(2)	(3)	(4)
Blue State (= 1)	−0.008** (0.003)	−0.008** (0.003)	−0.009* (0.004)	−0.010** (0.005)
Post-Fink (= 1 if Date ≥ 2022-01-01)	−0.013*** (0.003)	−0.013*** (0.003)	−0.014*** (0.003)	−0.015*** (0.003)
Blue State × Post-Fink (β_3)	0.020*** (0.005)	0.020*** (0.005)	0.021*** (0.005)	0.022*** (0.006)
log(1 + Market Cap)		−0.001 (0.003)	−0.001 (0.003)	−0.002 (0.003)
log(1 + CO ₂ Sc. 1+2)			−0.001 (0.002)	−0.0004 (0.002)
SBTi indicator				−0.002 (0.004)
Constant	0.009*** (0.002)	0.018 (0.036)	0.025 (0.041)	0.036 (0.043)
Sector fixed effects	No	No	No	Yes
Observations	228	228	214	214
R ²	0.037	0.037	0.039	0.048

Note: *p<0.1; **p<0.05; ***p<0.01. Dependent variable is event-level CAR(0,2). CR2 cluster-robust standard errors clustered on US state of headquarters (Head.Province, $G = 34$). Stars based on cluster-robust t-statistics with $df = G - 1$. Regression SEs are over event-level CAR outcomes and do not propagate market-model AR-estimation variance. Sample: US-headquartered events with a state classification, 2019-01-01 onwards (aligned with Fig. 5 and Extended Data Table 2). Pro-ESG and Neutral States vs Anti-ESG States as defined in the Methods. Post threshold: 1 January 2022 (approximating Fink’s January 2022 letter).



Extended Data Fig. 3: Robustness checks: Alternative modeling of abnormal returns
Note: Forest plot of the day-2 CAAR across specifications; thick and thin bars are 90% and 95% confidence intervals. Baseline: CAR(0,2), country benchmark index, 75-trading-day estimation window, no estimation–event buffer, 5% per-tail censoring of daily abnormal returns, 15% maximum missing share in estimation-window returns. Light blue: benchmark change (S&P 500, constant-mean returns). Purple: abnormal-return model (market-adjusted, constant $\beta = 1$, buy-and-hold). Dark blue: estimation window (100/250/500 days). Green: buffer window (8/16 days). Pink: censoring rate (0/1/10/15%). Dark red: maximum missing-return share (5/10%). Green (bottom): value-weighted CAAR. BHAR and value-weighted rows use percentile-bootstrap confidence intervals; all other rows use cross-period-independence standard errors.

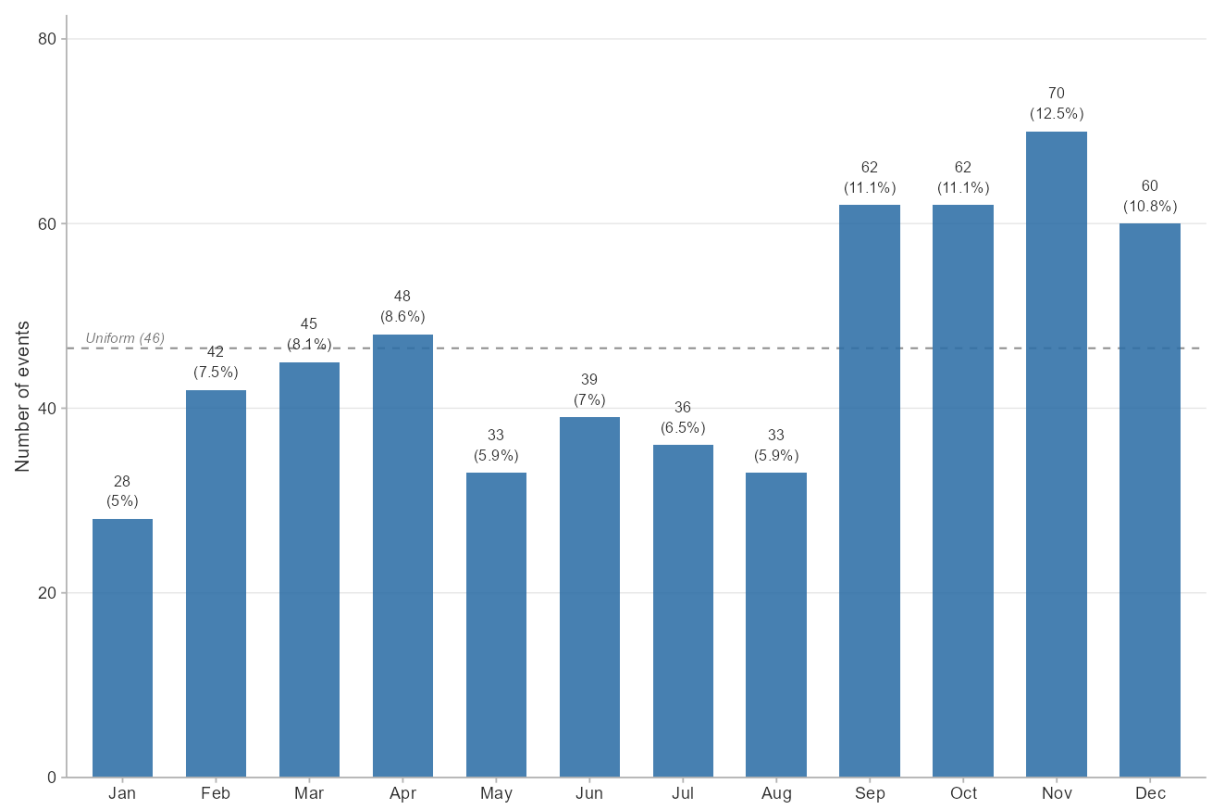
Supplementary Information

Modest and politically contingent stock market value for corporate climate pledges

Mathias Abitbol Renaud Coulomb

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Data description



Supplementary Figure 1: Distribution of commitment announcements by month

Notes: Number of retained commitment announcements by calendar month of release ($N = 558$ events). The dashed line is the expected count under a uniform distribution across months. Elevated counts in September–December coincide with the run-up to and holding of the annual COP meetings.

(a)

June 13, 2022



Helios Technologies Sets Goal of Achieving Net Zero GHG Emissions By 2050

SARASOTA, Fla.--(BUSINESS WIRE)-- [Helios Technologies, Inc.](https://www.helios technologies.com) (NYSE: HLIO) ("Helios" or the "Company"), a global leader in highly engineered motion control and electronic controls technology for diverse end markets, announced today it has set a long-term commitment to achieve net zero greenhouse gas emissions (GHG) by 2050 for its operated assets.

"We take our societal responsibility seriously and this newest commitment adds to our earlier recognition that Helios Technologies' environmental, social and governance (ESG) responsibilities are the living, operating principles by which we measure ourselves and shape our behavior on a daily basis," commented Josef Matosevic, President and Chief Executive Officer. "Initial efforts and steps are underway as we develop more detailed plans to reach this long-term commitment. At the Board level, we established an ESG Committee as a component of our governance structure to assist the Company in its oversight of corporate social responsibilities, significant public policy issues, health and safety, and climate-change related trends and other global ESG matters."

"We believe this is an important step forward in building a more sustainable company that can create long-term value for all stakeholders."

Helios has completed its first Scope 1 and Scope 2 greenhouse gas inventory and is working on detailed emission-reduction roadmaps for its major operated assets.

Read more about the Company's progress and follow its ESG journey at <https://helios technologies.com/esg>

About Helios Technologies

Helios Technologies is a global leader in highly engineered motion control and electronic controls technology for diverse end markets, including construction, material handling, agriculture, energy, recreational vehicles, marine, health, and wellness. Helios sells its products to customers in over 90 countries around the world. Its strategy for growth is to be the leading provider in niche markets, with premier products and solutions through innovative product development and acquisition. The Company has paid a cash dividend to its shareholders every quarter since becoming a public company in 1997. For more information please visit: www.helios technologies.com.

View source version on businesswire.com:

<https://www.businesswire.com/news/home/20220613005167/en/>

(b)

Nikon Group's Net-zero Targets Approved by Science Based Targets (SBT) initiative

March 22, 2024



DRIVING AMBITIOUS CORPORATE CLIMATE ACTION

TOKYO - Nikon Corporation (Nikon) received approval for its net-zero targets from the Science Based Targets (SBT) initiative. The Nikon Group has been setting a target of achieving carbon neutrality by FY2050. This time, the Nikon Group has set a new long-term target of achieving zero greenhouse gas (GHG) emissions^{*1} throughout the value chain following SBT initiative, which received approval from the SBT initiative. In addition, the GHG emissions reduction targets for FY 2030 (Near-Term Targets), which had previously been certified have been updated in line with the latest criteria set by the SBT initiative, and have been re-certified as the "1.5 degrees Celsius target."

Nikon will bring the target forward by 20 years for switching to 100% renewable energy for electricity used in our business activities, moving it from FY 2050 to FY 2030, to achieve the GHG emissions reduction targets.

The SBT Initiative is an initiative jointly established in 2015 by the United Nations Global Compact, WRI (World Resources Institute), and others to encourage companies to set science-based GHG reduction targets to limit the rise in global average temperatures due to climate change to 1.5°C above pre-industrial levels.

[SBT Initiative website](https://sciencebasedtargets.com)

Nikon Group's GHG emissions reduction targets approved by SBT initiative

- Net-Zero Target
Reach net-zero GHG emissions across the value chain by FY2050.
- Near-Term Targets
• Reduce absolute scope 1^{*2} and 2^{*3} GHG emissions 57% by FY2030 from a FY2022 base year.
• Reduce absolute scope 3^{*4} GHG emissions 25% by FY2030 from a FY2022 base year.

At the Nikon Group, in accordance with our Corporate Philosophy of Trustworthiness and Creativity, we meet society's expectations to win "trust" in each area of the environment, society/labor, and governance. Furthermore, we aim to contribute to the achievement of the realization of a sustainable society where humans and machines co-create seamlessly by "creating" new value through business activities. We continue to promote these efforts.

^{*1} Net-zero means reducing GHG emissions (Scope 1, 2, and 3) across the value chain by 90% and neutralizing remaining emissions in accordance with standards set by the SBT Initiative.

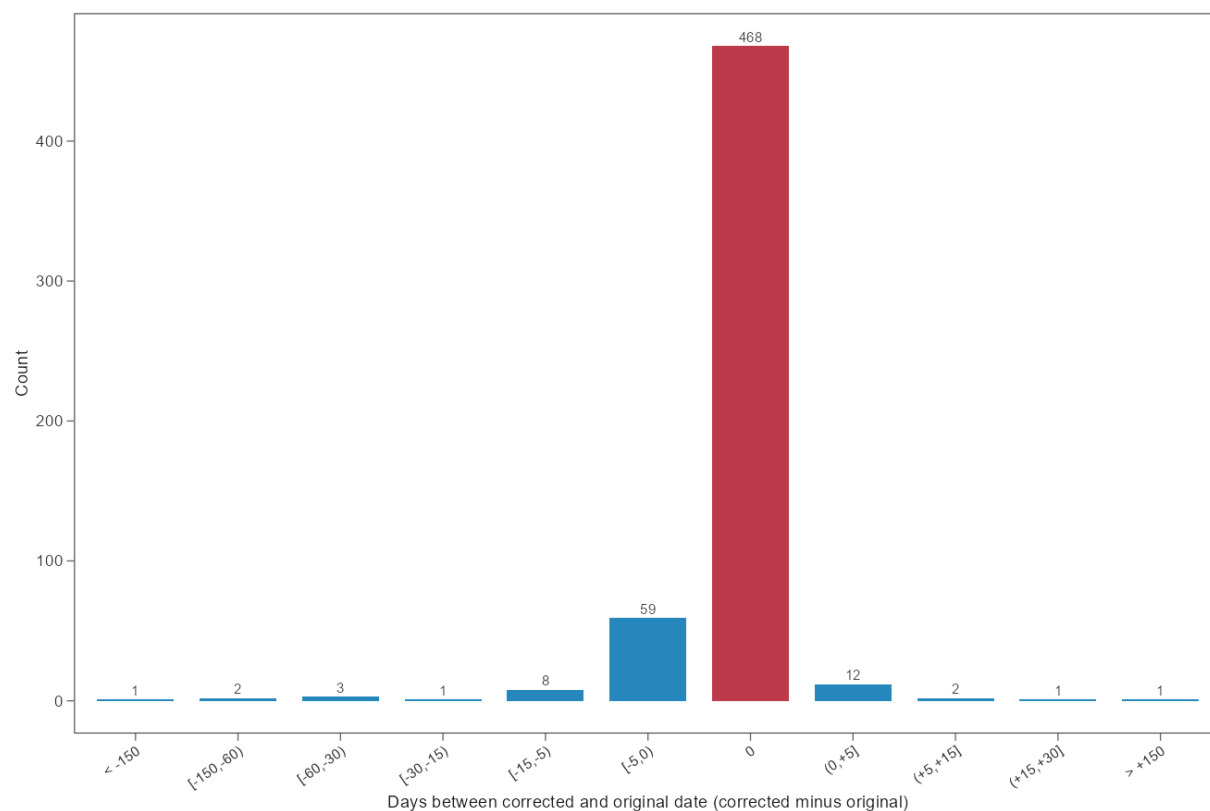
^{*2} Direct GHG emissions due to the use of fuel on site.

^{*3} Indirect GHG emissions from consumption of purchased electricity heat or steam.

^{*4} Other; indirect emissions of GHG emissions (excluding Scope 1 and 2).

Supplementary Figure 2: Examples of press releases announcing corporate climate commitments

Notes: Panel (a) shows an example of a press release announcing a new net-zero commitment, as recorded by the LSEG press aggregator (Helios Technologies, June 2022). Panel (b) shows an example of a press release announcing the validation of a target by the Science Based Targets initiative (Nikon, March 2024).



Supplementary Figure 3: Distribution of gaps between verified and recorded release dates

Notes: Distribution of the gap (in days) between the manually verified date on which the information became public and the date initially recorded by the LSEG press aggregator (corrected minus original; negative values indicate that the verified public date precedes the recorded date). The red bar marks exact matches.

Descriptive Statistics

Supplementary Table 1: Descriptive statistics of the PR sample

General characteristics							
Variable	n	Mean	SD	d1	d5	d9	N_{firm}
Release per firm	452	1.14	0.39	1.00	1.00	2.00	452
Market Capitalization (in B\$)	516	34.43	100.78	1.27	12.07	76.46	452
GhG Emissions, sc.1&2 (in Mt)	472	4.48	16.50	0.01	0.28	6.79	409
GhG Emissions int. (in tons/M\$)	471	304.30	887.37	3.07	29.68	620.66	408
Carbon Price (in €)	460	27.59	42.52	0.24	5.14	78.38	403
Carbon Price inc. FA (in €)	460	23.52	40.99	0.03	1.21	75.86	403

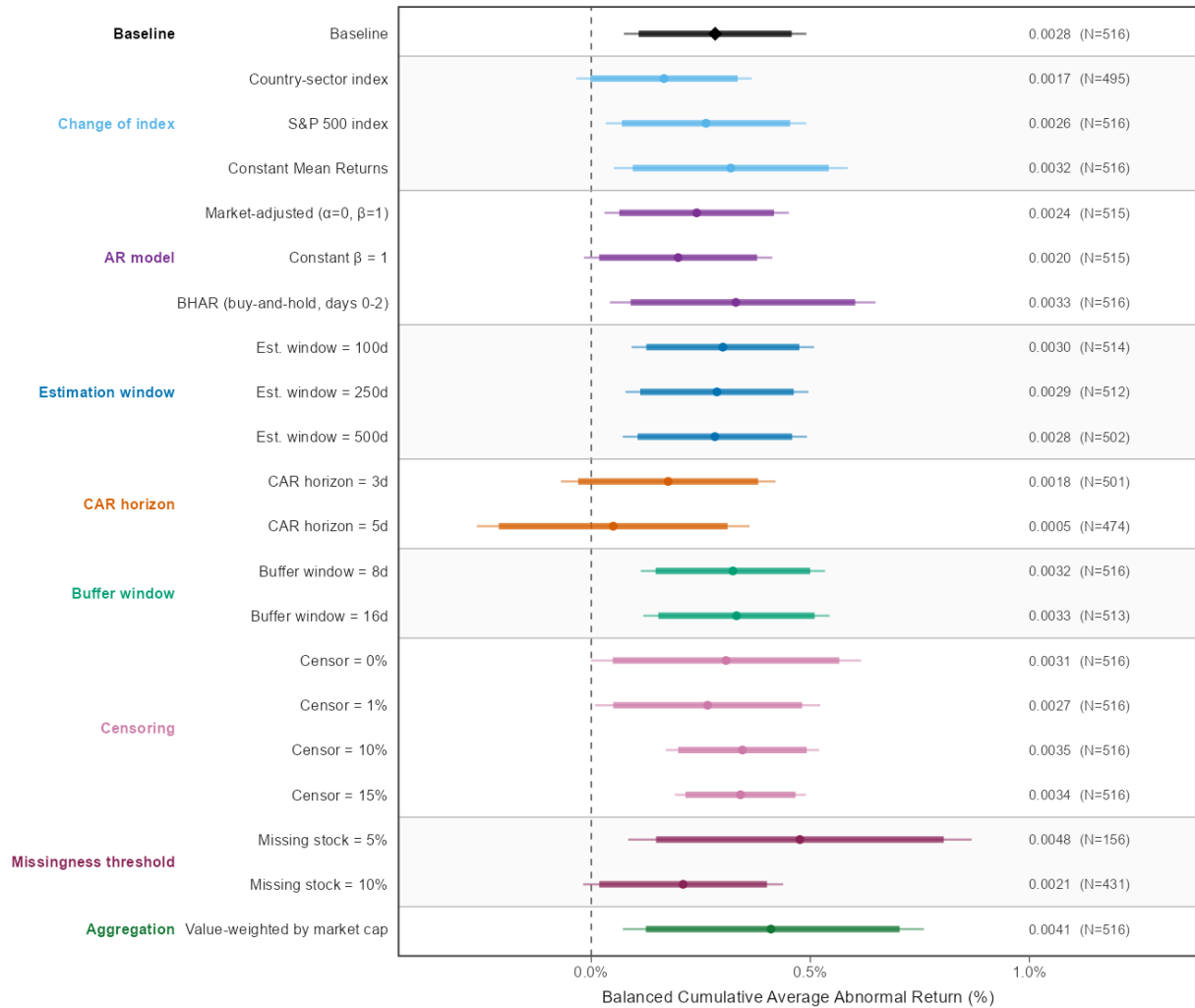
GHG Emissions Trend							
Metric	N_{firm}	Y-3	Y-2	Y-1	Y0	Y+1	Y+2
Sc.1 (MtCO ₂)	294	5.688	5.507	5.320	4.968	4.590	4.506
Av. Sc.1&2 (MtCO ₂)	289	5.960	5.848	5.640	5.336	5.064	4.942
Δ ind.-adj. Sc.1&2 (%)	287	0.252	-0.557	0.293	0.075	-0.123	0.343
Δ YoY Sc.1&2 (%)	287	—	-1.251	1.238	4.680	-0.431	1.503
Sc.1&2 int. (tCO ₂ /M\$)	250	387.595	360.072	361.977	364.045	291.330	267.877

Sectors				Regions			
Sector	n	pct	N_{firm}	Country	n	pct	N_{firm}
Finance & Tech	172	33.30	154	US	223	43.20	193
Customer Services	132	25.60	116	OECD, others	130	25.20	118
Industry & Utilities	115	22.30	101	EU	123	23.80	107
Energy & Mining	97	18.80	81	Non-OECD	40	7.80	34

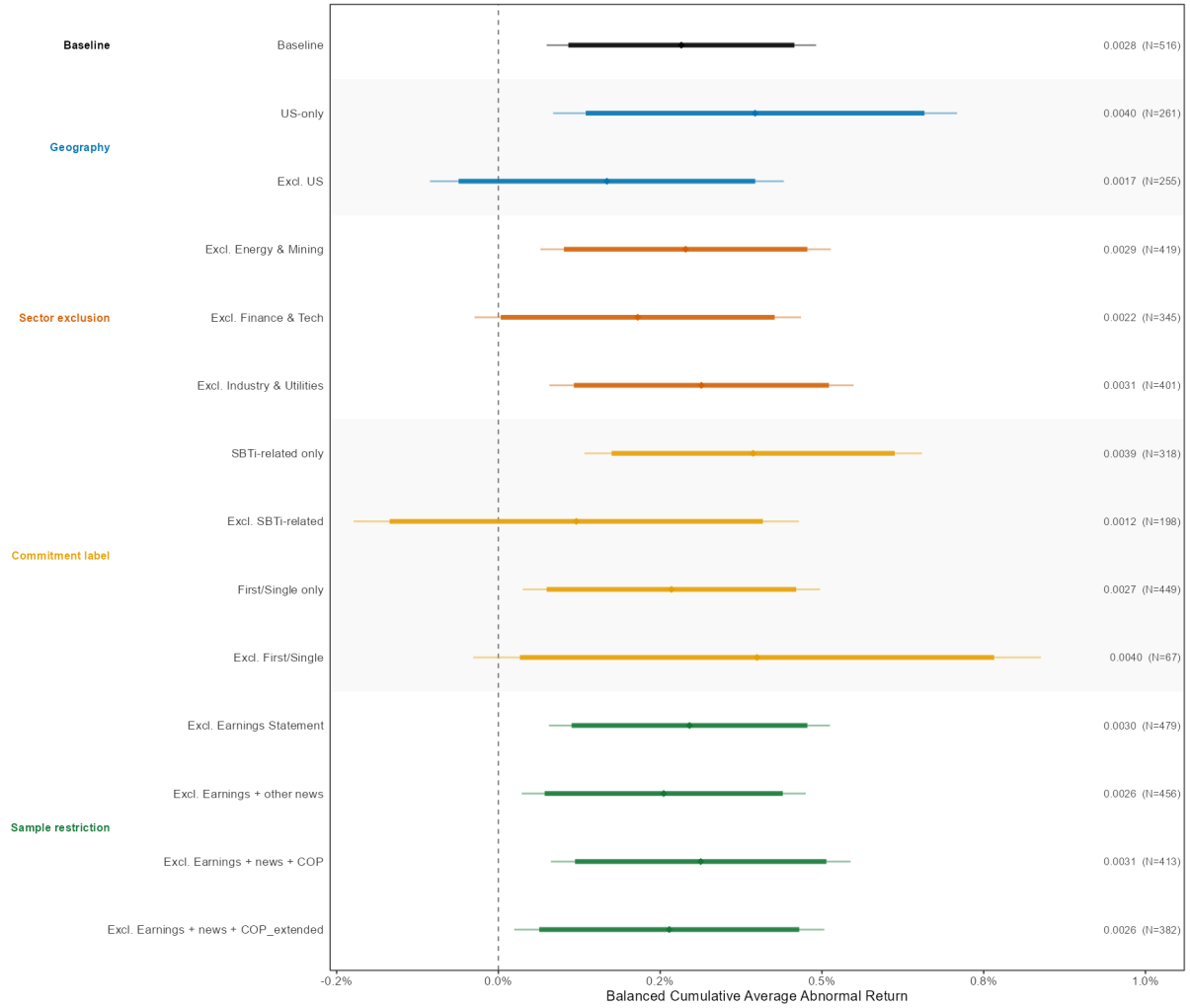
Type of label				Release Year			
Label	n	pct	N_{firm}	Year	n	pct	N_{firm}
No label	83	16.10	83	2017	3	0.6	3
SBTi	318	61.60	285	2018	—	—	—
<i>inc. commitment</i>	88	17.10	87	2019	11	2.1	11
<i>inc. submission</i>	37	7.20	37	2020	33	6.4	33
<i>inc. validation</i>	193	37.40	185	2021	170	32.9	163
GFANZ signatory	18	3.50	18	2022	147	28.5	145
Target modification	97	18.80	90	2023	152	29.5	150

Note: For each company, we select the first release within the sample. For the relevant variables, we restrict the sample to firms with no missing data over the window $[Y-3; Y+2]$. For sectors, countries, type of label and release years, n is the number of commitment events and N is the number of distinct companies. The emissions variables follow Bolton and Kacperczyk (2021) and Ilhan et al. (2021).

Balanced CAR Robustness

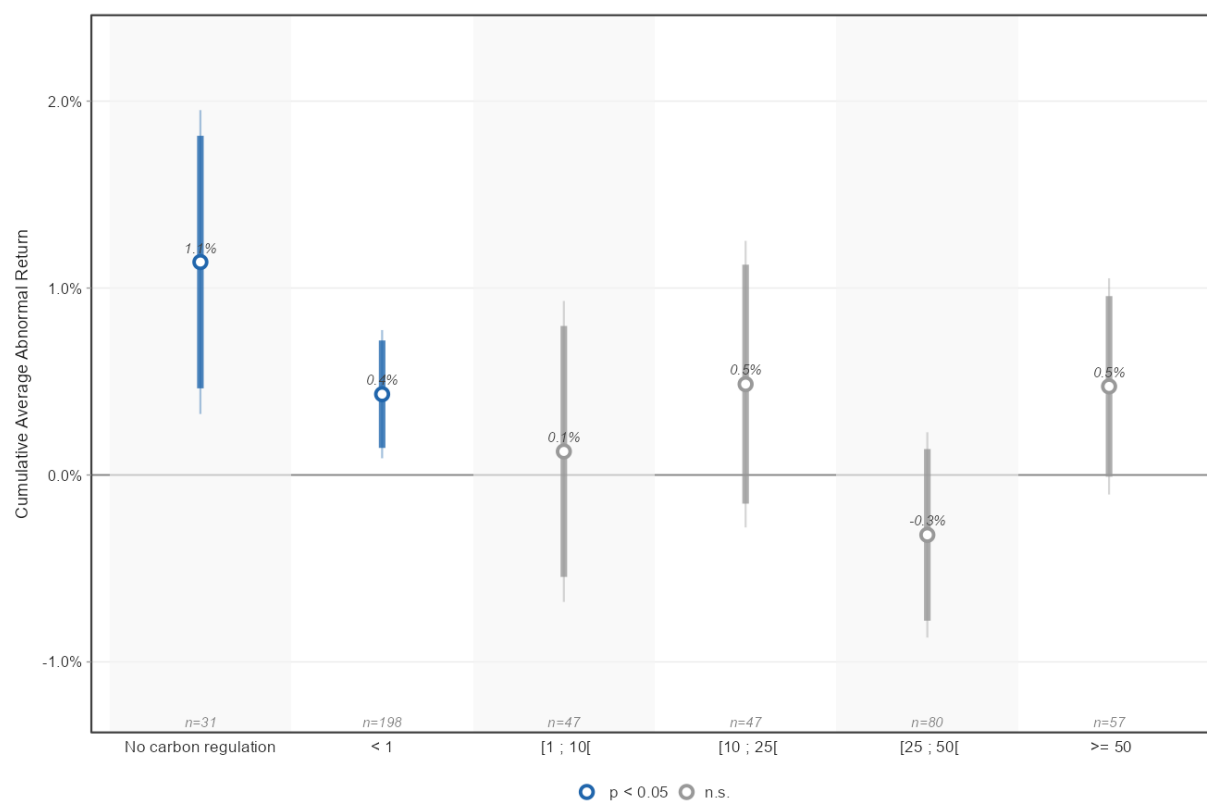


Supplementary Figure 4: Robustness checks under the balanced firm-level CAR rule
Notes: Same specifications as the main robustness forest (Extended Data Figure 3), but every headline point is re-estimated under the balanced firm-level CAR rule: per-event CAR(0,2) is the sum of daily abnormal returns with no missing day allowed, the specification CAAR is the cross-firm mean, and confidence intervals use Brown–Warner cross-sectional standard errors ($df = N - 1$). Thick and thin bars are 90% and 95% confidence intervals. BHAR and value-weighted rows keep their percentile-bootstrap intervals.



Supplementary Figure 5: Extended robustness checks under the balanced firm-level CAR rule

Notes: Same specifications as the extended robustness forest (Fig. 2 in the main text), re-estimated under the balanced firm-level CAR rule (complete event-window abnormal returns only; Brown–Warner cross-sectional standard errors, $df = N - 1$). Thick and thin bars are 90% and 95% confidence intervals.



Supplementary Figure 6: Day-2 CAAR by effective carbon rate, balanced firm-level CAR rule

Notes: Balanced-rule companion to main-text Figure 4. Day-2 CAAR by bin of the OECD Effective Carbon Rate including free allocations (€/tCO₂), under the balanced firm-level CAR rule (complete-case CAR(0,2); an event is dropped if any day in the cumulation window is missing), for the 459 of 558 events with a matched ECR and a complete CAR. Per-bin CAAR is the cross-firm mean with Brown–Warner cross-sectional standard errors ($df = N - 1$). Thick and thin bars are 90% and 95% confidence intervals. Bin sample sizes range from 31 (no carbon regulation) to 198 (<1 €). Confidence intervals are not adjusted for multiple comparisons across bins.

Carbone-price Regressions

Supplementary Table 2: Market reaction to mitigation announcement and effective carbon rate (OECD/ECR-coverage subsample)

	CAR(0,2)			
	(1)	(2)	(3)	(4)
Effective Carbon Rate, EMCR (log)	−0.002 (0.002)			
Effective Carbon Rate, EACR (log)		−0.002 (0.002)	−0.002 (0.002)	−0.002 (0.002)
CO2 Emissions, Sc. 1+2 (log)			−0.0004 (0.001)	−0.001 (0.001)
Share Committed (LSEG)				0.004 (0.006)
Constant	0.005*** (0.001)	0.005*** (0.001)	0.008 (0.007)	0.009 (0.007)
Discounting Free Allocation	No	Yes	Yes	Yes
Controlling for Carbon Emissions	No	No	Yes	Yes
Share Committed	No	No	No	Yes
Observations	422	422	422	422
R ²	0.004	0.004	0.004	0.004

Note: Each observation is a new net-zero commitment or an upward target revision. The analysis uses the PR sample restricted to OECD/ECR-covered complete cases (N = 422 of 558 PR events). Effective carbon rates are drawn from the OECD database; the 'Discounting Free Allocations' variable refers to an OECD ECR measure that adjusts for free allowances. Sector classification relies on the OECD/ISIC correspondence table, subsequently mapped to GICS sectors and industries using the UNEP FI ISIC/GICS crosswalk. The dependent variable is event-level CAR(0,2), computed as a complete-case sum of daily abnormal returns. Standard errors are CR2 cluster-robust SEs clustered at the country-of-incorporation level, matching the headline multivariate table; they do not propagate market-model AR-estimation variance.

Robustness Anti-ESG Analysis

Supplementary Table 3: Anti-ESG DiD: inference sensitivity

	CAR(0,2)					
	(1)	(2)	(3)	(4)	(5)	(6)
Blue State (= 1)	−0.008** (0.003)	−0.008* (0.004)	−0.008* (0.004)	−0.010** (0.005)	−0.010** (0.004)	−0.010* (0.005)
Post-Fink (= 1 if Date ≥ 2022-01-01)	−0.013*** (0.003)	−0.013* (0.006)	−0.013** (0.005)	−0.015*** (0.003)	−0.015* (0.006)	−0.015*** (0.006)
Blue State × Post-Fink (β_3)	0.020*** (0.005)	0.020** (0.006)	0.020*** (0.007)	0.022*** (0.006)	0.022*** (0.005)	0.022*** (0.007)
log(1 + Market Cap)				−0.002 (0.003)	−0.002 (0.003)	−0.002 (0.003)
log(1 + CO ₂ Sc. 1+2)				−0.0004 (0.002)	−0.0004 (0.001)	−0.0004 (0.002)
SBTi indicator				−0.002 (0.004)	−0.002 (0.005)	−0.002 (0.004)
Constant	0.009*** (0.002)	0.009* (0.004)	0.009*** (0.003)	0.036 (0.043)	0.036 (0.046)	0.036 (0.038)
Sector fixed effects	No	No	No	Yes	Yes	Yes
SE scheme	HQ-state CR2	Inc.-country CR2	HC1	HQ-state CR2	Inc.-country CR2	HC1
Clusters G	34	7	—	33	7	—
Observations	228	228	228	214	214	214
R ²	0.037	0.037	0.037	0.048	0.048	0.048

Note: Columns (1)–(3) re-estimate the baseline DiD and columns (4)–(6) the full-controls specification of the main anti-ESG table under three inference schemes: CR2 cluster-robust SEs clustered by state of headquarters (primary), CR2 clustered by country of incorporation, and heteroskedasticity-robust HC1. Cluster-robust stars use t -statistics with $df = G - 1$; HC1 stars use $df = N - k$. Coefficients are identical across schemes by construction; only the SEs and stars vary.

Supplementary Table 4: Anti-ESG DiD robustness: treatment date and state ambiguity

	CAR(0,2)		
	Baseline (Post $\geq$ 2022-01-01)	Post $\geq$ 2021-06-01 (Texas SB 13/19)	Excl. VA & NC
	(1)	(2)	(3)
Blue State (= 1)	−0.010** (0.005)	−0.008 (0.007)	−0.011** (0.005)
Post-Fink / Post-Texas	−0.015*** (0.003)	−0.010*** (0.004)	−0.015*** (0.004)
Blue $\times$ Post	0.022*** (0.006)	0.011 (0.007)	0.022*** (0.007)
log(1 + Market Cap)	−0.002 (0.003)	−0.001 (0.003)	−0.001 (0.003)
log(1 + CO ₂ Sc. 1+2)	−0.0004 (0.002)	0.0001 (0.002)	−0.001 (0.002)
SBTi indicator	−0.002 (0.004)	−0.001 (0.004)	−0.004 (0.004)
Constant	0.036 (0.043)	0.030 (0.038)	0.032 (0.045)
Sector FE	Yes	Yes	Yes
State cluster (G)	33	33	31
Observations	214	214	202
R ²	0.048	0.016	0.045

Note: CR2 cluster-robust SEs (Head_Province) in parentheses. Col. (1): reference specification (full controls; identical to column (4) of Extended Data Table 3), Post = $\mathbb{1}[\text{Date} \geq 2022-01-01]$. Col. (2): Post re-defined as $\mathbb{1}[\text{Date} \geq 2021-06-01]$ to capture Texas SB 13 (signed June 2021), which predates the Fink letter. Col. (3): Virginia and North Carolina (both had Democratic governors at the time of key legislation) excluded from the Anti-ESG bucket to test classification sensitivity.

Supplementary Note 1: Abnormal returns and CAAR estimation

The measure of abnormal returns related to firms' commitment. In the Efficient Market Hypothesis (EMH) [46], at the market equilibrium, the expected abnormal return should be 0: $\mathbb{E}(AR_{i,t+1} | \Omega_t) = 0$, Ω_t information set at date t . Hence, any abnormal return must be event-induced, i.e. derived from some new information that made the market depart from its original valuation of a firm stock. The abnormal return in a specific date (or succession of dates) can thus be associated to the specific impact of the newly arrived set of information.

Our first objective is to measure the market's reaction to commitment release. We follow the method set up in [47] and [48], and updated in [49] and [50] by constructing firm abnormal returns following the date of the announcement¹. For each commitment, we first estimate the relationship between the firm's return and that of the country market over a period of 75 trading days (roughly three and a half months) before the event window. Country market return is estimated through the use of a representative index. We then predict the firm's returns on an event window around the day of the event using the estimated market model and the observed market returns. Finally, we define abnormal returns as the difference between the observed and predicted returns. These abnormal returns are then averaged across firms. Supplementary Fig. 7 summarises the timing conventions.

In detail, let t_0 denote the announcement day of event i , and let $\tau_A = t_0 - 83$, $\tau_B = t_0 - 8$ and $\tau_C = t_0 + 15$ be the three dates (in trading days) marked in Supplementary Fig. 7: the calibration window runs over $[\tau_A, \tau_B]$ and the abnormal-return window over $[\tau_B, \tau_C]$. We first run the following regression for each firm and each commitment² to calibrate the market model (based on the specific market index of the country of exchange of the firm's main equity) in order to calibrate the firm i 's α_i and β_i parameters:

$$R_{i,t} = \alpha_i + \beta_i R_{c,t}^m + \varepsilon_{i,t}, \quad t \in [\tau_A, \tau_B] \quad (\text{S1})$$

where the calibration window $[\tau_A, \tau_B]$ comprises the 75 trading days preceding the abnormal-return window³, $R_{i,t}$ is the firm i stock return on day t , $R_{c,t}^m$ is the market return for the country of exchange $c = c(i)$ ⁴ on day t and $\varepsilon_{i,t}$ is the zero-mean disturbance term, assumed normally distributed and uncorrelated with the market return. Using the market model has become the gold standard in the past twenty years⁵.

¹Following a broad consensus in this literature, we will use a daily event-study. The method was initially conducted at a monthly frequency, but it has become standard to measure it on a daily basis [50], because of the lack of precision of the initial method. Nevertheless, intra-day event-studies are usually deemed too sensitive to high-frequency trading, besides the data being less easily accessible. Nor is the abnormal-return design specific to financial economics: it is widely used across applied economics, for instance in the *Journal of the European Economic Association* [51], the *Journal of Public Economics* [52], and the *Journal of Law, Economics, and Organization* [53].

²Technically speaking, some firms have several commitment releases on the timespan of the study. For simplicity, in what follows, we will neglect this aspect, and refer without loss of generality to the i index indistinctly as an index for commitments and firms.

³In the baseline there is no buffer between the calibration window and the abnormal-return window: they meet at $\tau_B = t_0 - 8$. Robustness checks insert an 8- or 16-day buffer; the headline CAR(0,2) is essentially unchanged (see the robustness forest).

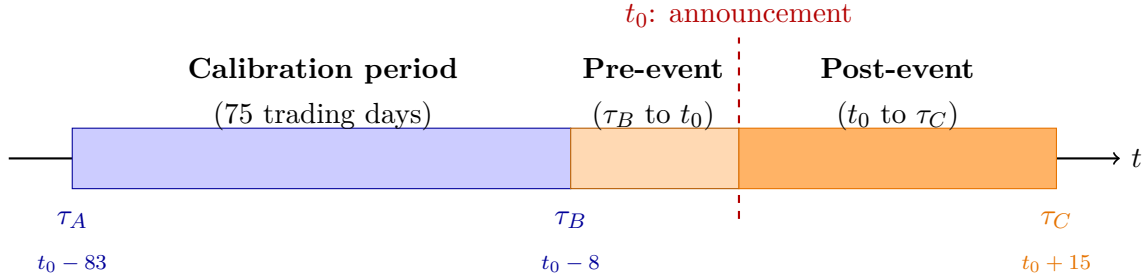
⁴Each firm i is associated to one single exchange country depending on the place where its main equity is being traded.

⁵It has been preferred over the CAPM technique because of the decisive critique from [54], and because

In turn, it allows us to compute the abnormal return $AR_{i,\tau}$ for each firm i on every trading day τ of the abnormal-return window $[\tau_B, \tau_C]$ around the event day t_0 , as an out-of-sample prediction:

$$\widehat{AR}_{i,\tau} = R_{i,\tau} - \left(\hat{\alpha}_i + \hat{\beta}_i R_{c,\tau}^m \right), \quad \tau \in [\tau_B, \tau_C] \quad (\text{S2})$$

where $\hat{\alpha}_i$ and $\hat{\beta}_i$ are estimated through the market model for each firm i . The abnormal return is the disturbance term of the market model calculated on an out-of-sample basis. We winsorise (clamp, not drop) 5% of the distribution on each side for each different day within the event study; robustness checks vary this censoring rate between 0% and 15%.



Supplementary Figure 7: Event-study time-line. Trading-day indexing relative to the announcement day t_0 (horizontal axis not to scale). Three dates organise the design: the calibration (estimation) window runs over $[\tau_A, \tau_B]$, with $\tau_A = t_0 - 83$ and $\tau_B = t_0 - 8$ (75 trading days); abnormal returns are computed out of sample over $[\tau_B, \tau_C]$, with $\tau_C = t_0 + 15$; the headline CAR cumulates days t_0 to $t_0 + 2$; and the figures display $t_0 \pm 5$ days. The market-model parameters $(\hat{\alpha}_i, \hat{\beta}_i)$ are estimated on the calibration period and used to construct out-of-sample abnormal returns over the pre- and post-event windows. In the baseline there is no buffer between the calibration window and the abnormal-return window (they meet at τ_B), so the abnormal return plotted at τ_B is mechanically attenuated; the headline CAR(0,2) is unaffected. Robustness checks insert an 8- or 16-day buffer between the calibration window and τ_B .

These abnormal returns then allow for an aggregation along two dimensions, across days and across firms, where two distinct logics coexist — the balanced and AAR-cumulate series (the butterfly figures additionally overlay an available-day variant):

(i) *Firm-level (“balanced”) aggregation*: $AR \rightarrow CAR_i \rightarrow CAAR$. Each event’s abnormal returns are first cumulated into its own $\widehat{CAR}_i(t_1, t_2)$; the event enters only if $\widehat{AR}_{i,\tau}$ is observed on *every* day $\tau \in [t_1, t_2]$ (complete cases). The CAAR is then the cross-event mean,

$$\widehat{CAAR}^{bal}(t_1, t_2) = \frac{1}{N_{bal}} \sum_{i=1}^{N_{bal}} \widehat{CAR}_i(t_1, t_2), \quad (\text{S3})$$

it is deemed more sophisticated than simply assuming constant return over the event window; indeed, it allows us to estimate each company’s expected return in a much more powerful way, while removing any market-wide shock. On the contrary, [55] found it doubtful that the use of any additional factor beyond market returns brings any further insight.

where

$$\widehat{CAR}_i(t_1, t_2) = \sum_{\tau=t_1}^{t_2} \widehat{AR}_{i,\tau} \quad (\text{S4})$$

with Brown–Warner cross-sectional standard errors, $SE = sd(\widehat{CAR}_i)/\sqrt{N_{bal}}$.

(ii) *Calendar-day (“AAR-cumulate”) aggregation:* $AR \rightarrow AAR_\tau \rightarrow CAAR$. Abnormal returns are first averaged across events day by day, over the N_τ events with an observed abnormal return on day τ , and the daily averages are then cumulated:

$$\widehat{AAR}_\tau = \frac{1}{N_\tau} \sum_{i=1}^{N_\tau} \widehat{AR}_{i,\tau} \quad \text{and} \quad \widehat{CAAR}(t_1, t_2) = \sum_{\tau=t_1}^{t_2} \widehat{AAR}_\tau, \quad (\text{S5})$$

with cross-period-independence (CPI) standard errors, $SE = \sqrt{\sum_{\tau=t_1}^{t_2} \widehat{Var}(\widehat{AAR}_\tau)}$.

The two logics differ only in how missing firm-days are handled: the balanced rule drops any event with a missing day in the cumulation window ($N_{bal} \leq N_\tau$), while the AAR-cumulate rule uses every available event-day observation.

We compute CAARs for the full sample and for various subsamples—excluding specific countries, sectors, press-release types, or event windows—for our robustness checks, as indicated in the main text and in the notes to the figures and tables.

Supplementary Note 2: Power and equivalence bounds for the certification-premium null

The main text reports that the absence of a certification premium is informative rather than underpowered. This note details the two quantities behind that claim, the minimum detectable effect (MDE) and the equivalence bound, both computed from the label-level CAARs.

Let $\hat{C}_v$ and $\hat{C}_u$ denote the day-2 CAARs for SBTi-validated targets ($N_v = 193$) and unlabelled new targets ($N_u = 85$), estimated under the AAR-cumulate rule with cross-period-independence standard errors s_v and s_u (Eq. S5). The certification contrast and its standard error are

$$\hat{\delta} = \hat{C}_v - \hat{C}_u, \quad s_{\delta} = \sqrt{s_v^2 + s_u^2}, \quad (\text{S6})$$

treating the two label groups as independent cross-sections. In the analysis sample, $\hat{\delta} = 0.23$ pp with $s_{\delta} = 0.32$ pp, giving the 95% confidence interval $\hat{\delta} \pm z_{0.975} s_{\delta} = [-0.39, +0.86]$ pp.

Minimum detectable effect. For a two-sided test of $H_0: \delta = 0$ at the 5% level, the smallest true premium detectable with 80% power is

$$\text{MDE}_{80} = (z_{0.975} + z_{0.80}) s_{\delta} = 2.80 \times s_{\delta} = 0.89 \text{ pp}. \quad (\text{S7})$$

A certification premium of that size or larger would thus have been detected with at least 80% probability; the observed contrast is barely a quarter of it.

Equivalence bound. We complement the power calculation with a two-one-sided test. For a margin $\Delta > 0$, the equivalence bound rejects the hypothesis of a non-negligible premium, $H_0: |\delta| \geq \Delta$, at the 5% level whenever the 90% confidence interval for δ lies entirely within $(-\Delta, \Delta)$. The smallest margin for which this rejection obtains is

$$\Delta^* = |\hat{\delta}| + z_{0.95} s_{\delta} = 0.75 \text{ pp}, \quad (\text{S8})$$

so the data reject, at the 5% level, any announcement-stage certification premium larger than 0.75 percentage points.

Caveats. Both quantities use normal critical values, an accurate approximation at these cell sizes. They inherit the assumptions of the underlying standard errors: independence of abnormal returns across label groups (label groups are non-overlapping sets of events, but events overlap in calendar time, so residual cross-sectional correlation would understate s_{δ}), and, as throughout the paper, no propagation of market-model estimation variance from the daily abnormal-return step. The bounds concern the SBTi-validated-versus-unlabelled contrast only and are not adjusted for multiple comparisons across the other label cells of Extended Data Fig. 2.

Supplementary Note 3: Matching the carbon price

There are three dimensions that require a matching process between the OECD Effective Carbon Rates database and the PR sample: country, sector, and year. 67 observations are excluded from the binned carbon-price figure due to missing country ECR data or missing sector-level entries in the matching table.

Country dimension. The matching covers most countries in our sample. We use country of incorporation as a proxy for the relevant climate pricing scheme.

Year dimension. The OECD ECR database provides two cross-sections: 2018 and 2021. We match each observation to the nearest available year: commitment years 2017–2020 are matched to the 2018 ECR values, and commitment years 2021–2023 are matched to the 2021 values. Our sample spans six years (2017, 2019, 2020, 2021, 2022, 2023). No retained event occurs in 2018.

Sector dimension. The sectoral matching is performed via a correspondence table that maps each firm’s GICS industry classification to one of 20 OECD ECR user categories, following the OECD/ISIC correspondence table provided by the OECD, subsequently mapped to GICS sectors and industries using the UNEP FI ISIC/GICS crosswalk [56]. The most populated OECD categories in our sample are COMMPUB (commercial and public services, 275 firms), MACHINE (machinery and equipment, 101), CHEMICAL (67), ELEC (electricity producers, 41), FOODPRO (33), MINING (30), and WOODPRO (28). Several specific mappings deserve attention. Firms in *Oil, Gas & Consumable Fuels* are mapped to up to four different categories depending on their primary activity: ADJDISTLOSS (gas distribution), DOMESAIR (fuel for air transport), NONROAD (pipeline transportation), or COMMPUB (commercial services), depending on the granular ISIC Code. Electric utilities, independent power producers, and multi-utilities are uniformly mapped to ELEC. The mining sector is split between MINING and NONFERR.

Missing values. Of the 558 observations in our sample, 67 could not be matched to an ECR value and are excluded from the carbon pricing analysis. These unmatched observations fall into two categories.

The first and largest category comprises 49 observations whose country of incorporation is entirely absent from the OECD ECR database. This concerns ten countries: Taiwan (11 observations), Singapore (9), United Arab Emirates (8), Bermuda (7), Hong Kong (7), Curacao (2), Jersey (2), Cayman Islands (1), Liberia (1), and Malta (1). These are jurisdictions not covered by the OECD framework.

The second category comprises 18 observations for which the country is in principle covered by the OECD but the specific country×sector combination is absent from the ECR microdata. These cases arise either because the OECD does not compute an ECR for that sector in that country, or because the firm’s `user_OECD` code could not be determined from the matching table (`user_OECD = NA`). For the NA user cases, the underlying GICS industries (online services, business support services, paper packaging, passenger ground transport, UK investment trusts, and IT services) do not map unambiguously to any OECD energy-user category and were left unmatched in the correspondence table. The affected observations are: United States×NA (8), Switzerland×MINING (3), United Kingdom×NA (3), Australia×HEAT (1), France×NA (1), Israel×CHEMICAL (1), Japan×NA (1).

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